Course: Artificial Intelligence and Machine Learning Code: 20CS51I

WEEK- 2 Machine Learning

Machine Learning: Fundamentals

Machine learning (ML) is a type of artificial intelligence (AI) that allows software applications to become more accurate at predicting outcomes without being explicitly programmed to do so. Machine learning algorithms use historical data as input to predict new output values.

(OR)

Machine learning is a subset of AI, which enables the machine to automatically learn from data, improve performance from past experiences, and make predictions.

(OR)

Machine Learning is the field of study that gives computers the capability to learn without being explicitly programmed. ML is one of the most exciting technologies that one would have ever come across. As it is evident from the name, it gives the computer that makes it more similar to humans: The ability to learn. Machine learning is actively being used today, perhaps in many more places than one would expect.

Machine learning types

1. Supervised Machine Learning

As its name suggests, Supervised machine learning is based on supervision. It means in the supervised learning technique, we train the machines using the "labelled" dataset, and based on the training, the machine predicts the output.

Supervised learning works on labelled data. Each input data has a corresponding labelled output. The goal of supervised machine learning is to learn a mapping from the input to the output. The input data is called attributes, features or predictors. This output variable is also called response variable or target variable.

For example, the problem of building a utility for predicting the selling price of the car. The dataset is shown below:

Brand	Car Model	Car Registration Year	Miles driven	Price($)
Nissan	Sunny	2010	90000	10000
Toyota	Corolla	2009	45000	25000
Honda	Brio	2015	70000	18000
Chevrolet	Cruze	2013	60000	30000
Nissan	Terrano	2018	12000	40000
Honda	City	2017	25000	45000
.....				
.....				

Input Variables (X) Output Variable (Y)

From the given dataset, the machine learning algorithm learns the mapping from input variables to output variable. This learning is represented in the form of a model. When a new instance is given to the model as shown below, it can predict its output value.

Brand	Car Model	Car Registration Year	Miles driven	Price($)
Honda	City	2012	15000	?

Some of the other examples of supervised learning:

Given an email defined by its collection of phrases(X), predict if the mail is a spam(Y).

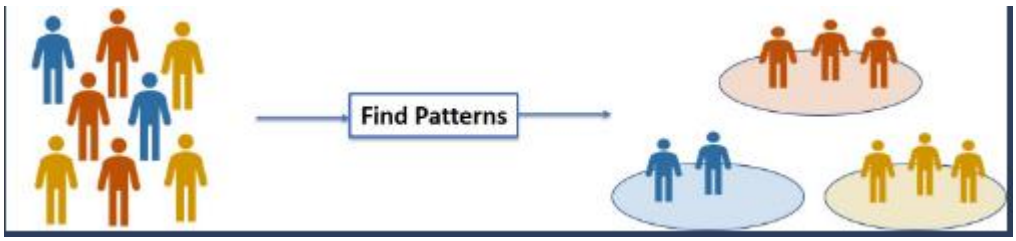
Given a medical brain scan image (X), predict if the patient has tumour(Y).

2. Unsupervised Machine Learning

unsupervised learning algorithm is to group or categories the unsorted dataset according to the similarities, patterns, and differences. Machines are instructed to find the hidden patterns from the input dataset.

Unsupervised machine learning has no explicitly defined output. The idea is to discover knowledge or structure in the data.

For example, an online retailer will have data about all items that the customers purchased. Unsupervised learning algorithms can be applied on this data to group customers based on their buying patterns.

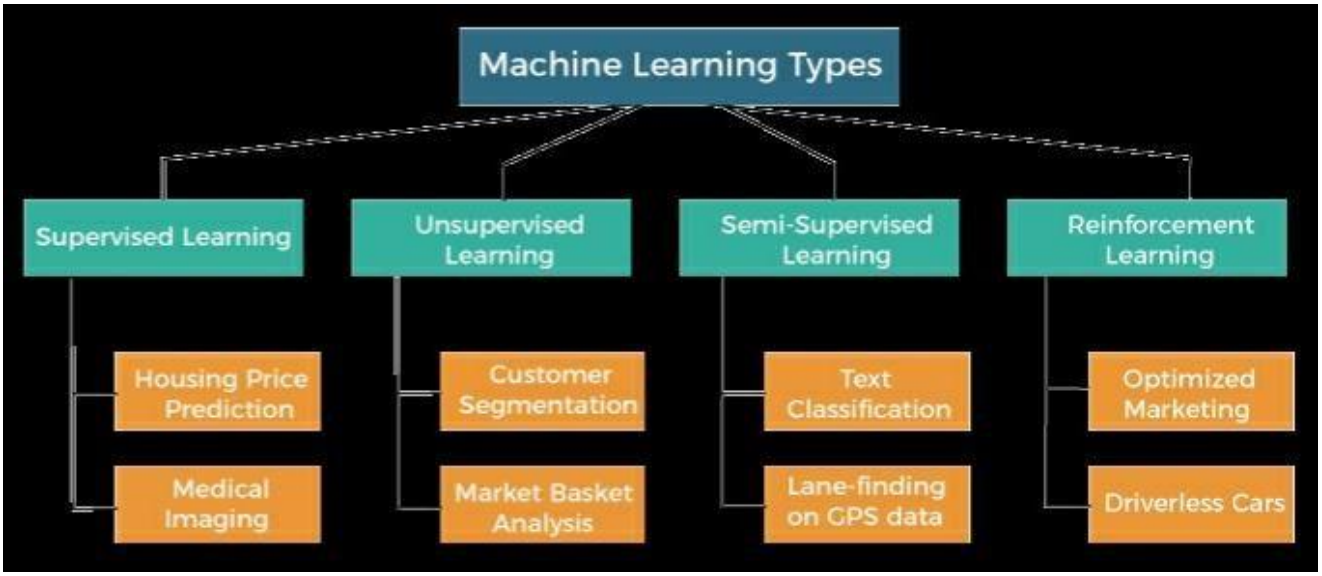


Grouping new articles based on topics like sports, politics, business etc. is another example of unsupervised learning.

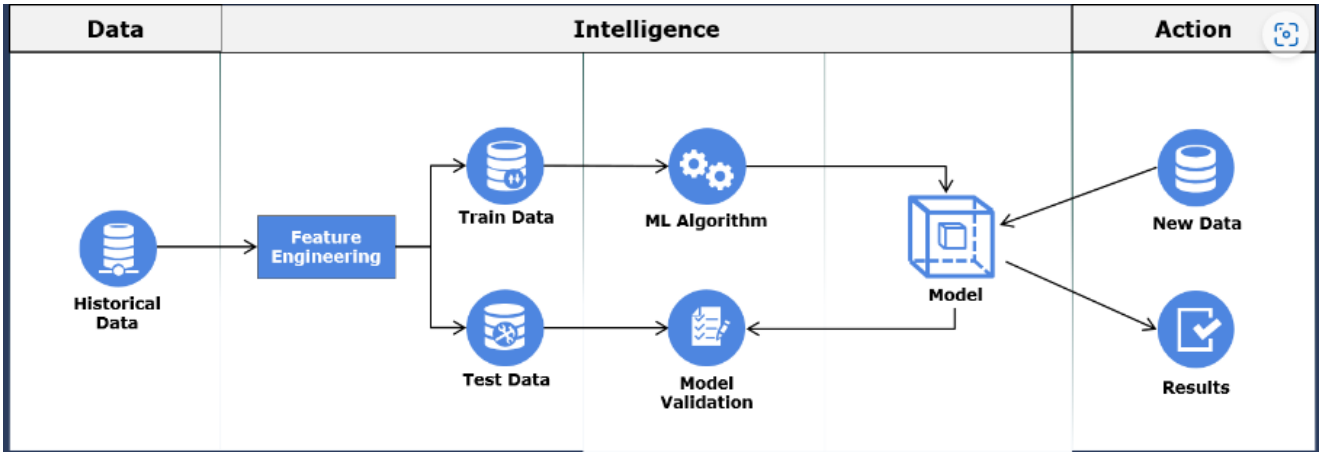
This task of discovering inherent clusters or groups in the data is known as Clustering

3. Reinforcement Learning

Reinforcement learning works on a feedback-based process, in which an AI agent (A software component) automatically explore its surrounding by hitting & trail, taking action, learning from experiences, and improving its performance.



Machine Learning Process



The diagram above illustrates the Machine Learning process.

In this process, first relevant data is gathered then is cleaned and transformed through a process called Feature Engineering. During the process of Feature Engineering, handling missing value, handling outliers, creating new features out of existing ones are some of the common tasks performed.

After feature engineering, the data is split into Train Data and Test Data. The Train Data is used for training the machine learning model. Once the model is built, it is validated against the Test Data for accuracy. This accuracy helps us in estimating the performance on previously unseen data. If the model performance on both Train and Test Data is satisfactory, the model may be deployed.

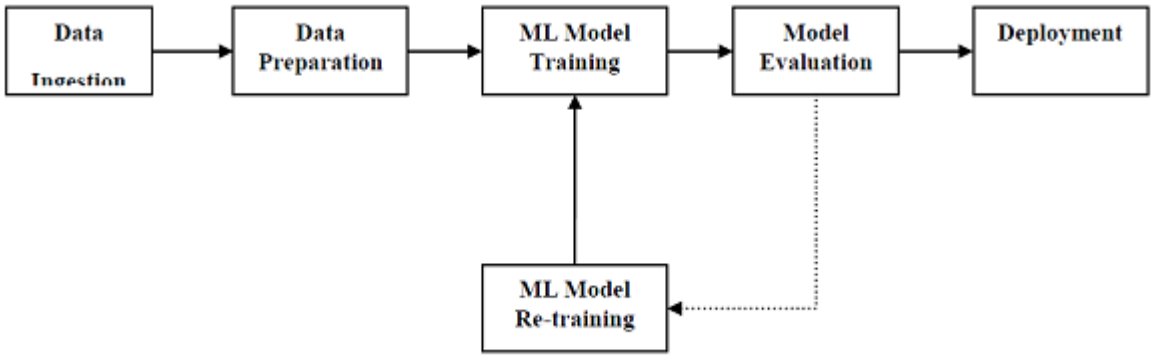
Once deployed, the model makes predictions on new data ; these predictions/insights are used to take business decisions

Machine learning workflow

In order to execute and produce results successfully, a machine learning model must automate some standard workflows. The process of automate these standard workflows can be done

with the help of Scikit-learn Pipelines. From a data scientist's perspective, pipeline is a generalized, but very important concept. It basically allows data flow from its raw format to some useful information.

The working of pipelines can be understood with the help of following diagram −



The blocks of ML pipelines are as follows −

Data ingestion − As the name suggests, it is the process of importing the data for use in ML project. The data can be extracted in real time or batches from single or multiple systems. It is one of the most challenging steps because the quality of data can affect the whole ML model.

Data Preparation − After importing the data, we need to prepare data to be used for our ML model. Data preprocessing is one of the most important technique of data preparation.

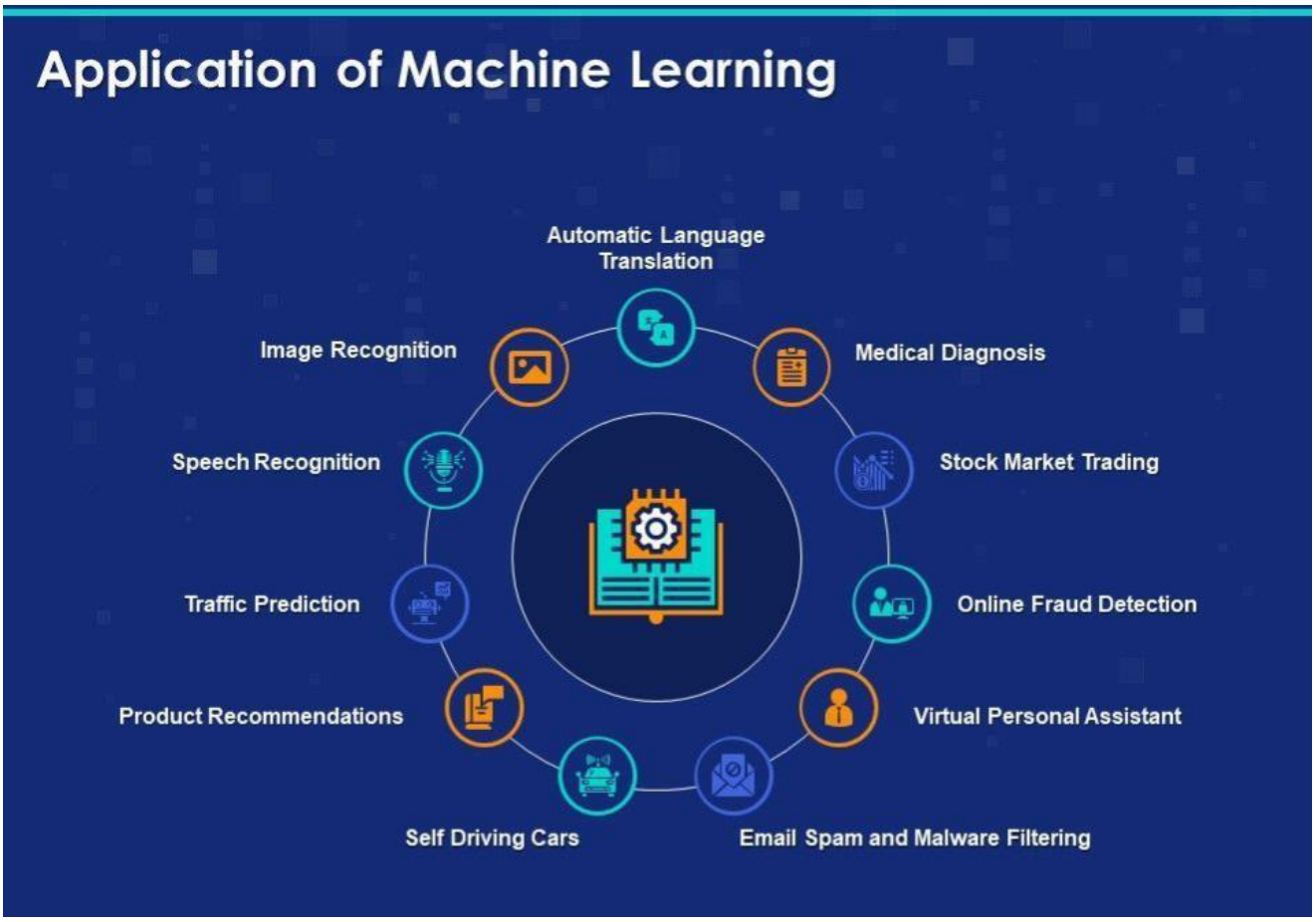
ML Model Training − Next step is to train our ML model. We have various ML algorithms like supervised, unsupervised, reinforcement to extract the features from data, and make predictions.

Model Evaluation − Next, we need to evaluate the ML model. In case of AutoML pipeline, ML model can be evaluated with the help of various statistical methods and business rules.

ML Model retraining − In case of AutoML pipeline, it is not necessary that the first model is best one. The first model is considered as a baseline model and we can train it repeatably to increase model's accuracy.

Deployment − At last, we need to deploy the model. This step involves applying and migrating the model to business operations for their use.

Machine learning Applications



4. Challenges in ML

- ✓ Inadequate Training Data
- ✓ Poor quality of data
- ✓ Monitoring and maintenance
- ✓ Getting bad recommendations
- ✓ Lack of skilled resources
- ✓ Process Complexity of Machine Learning
- ✓ Data Bias
- ✓ Slow implementations and results

Building model steps involved

Step 1: Collect Data

Given the problem you want to solve, you will have to investigate and obtain data that you will use to feed your machine.

Step 2: Prepare the data

This is a good time to visualize your data and check if there are correlations between the different characteristics that we obtained.

You must also separate the data into two groups: one for training and the other for model evaluation which can be divided approximately in a ratio of 80/20 but it can vary depending on the case and the volume of data we have.

At this stage, you can also pre-process your data by normalizing, eliminating duplicates, and making error corrections.

Step 3: Choose the model

you will use algorithms of classification, prediction, linear regression, clustering, i.e. k-means or K-Nearest Neighbor, Deep Learning, i.e Neural Networks, Bayesian, etc.

There are various models to be used depending on the data you are going to process such as images, sound, text, and numerical values.

In the following table, we will see some models and their applications that you can apply in your projects:

Model	**Applications**
Logistic Regression	Price prediction
Fully connected networks	Classification
Convolutional Neural Networks	Image processing
Recurrent Neural Networks	Voice recognition
Random Forest	Fraud Detection
Reinforcement Learning	Learning by trial and error
Generative Models	Image creation
K-means	Segmentation

Model	Applications
k-Nearest Neighbors	Recommendation systems
Bayesian Classifiers	Spam and noise filtering

Step 4 Train your machine model

You will need to train the datasets to run smoothly and see an incremental improvement in the prediction rate.

Step 5: Evaluation

You will have to check the machine created against your evaluation data set of your already trained model.

If the accuracy is less than or equal to 50%, that model will not be useful.

If you reach 90% or more, you can have good confidence in the results that the model gives you.

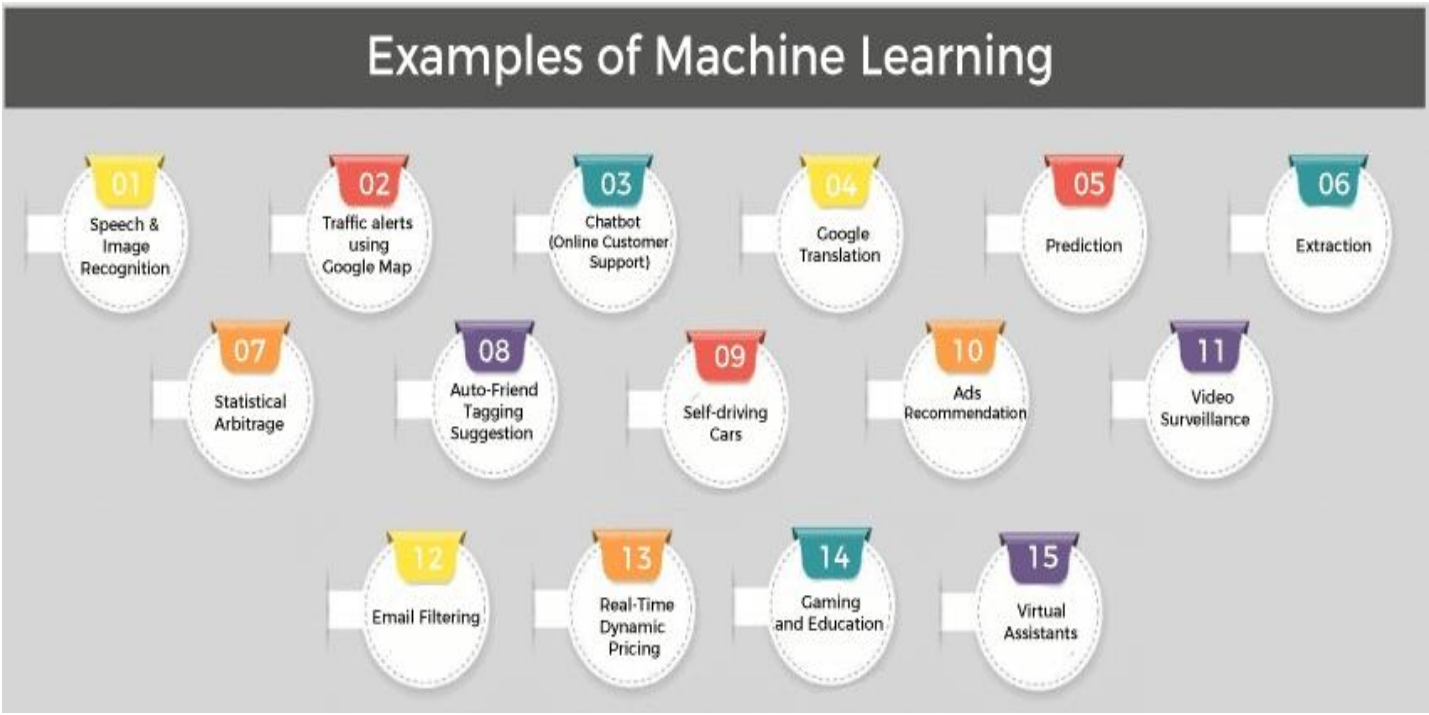
Step 6: Parameter Tuning

If during the evaluation you did not obtain good predictions, you must return to the training step before making a new configuration of parameters in your model.

Step 7: Prediction or Inference

You are now ready to use your Machine Learning model inferring results in real-life scenarios.

Examples of ML in Daily life

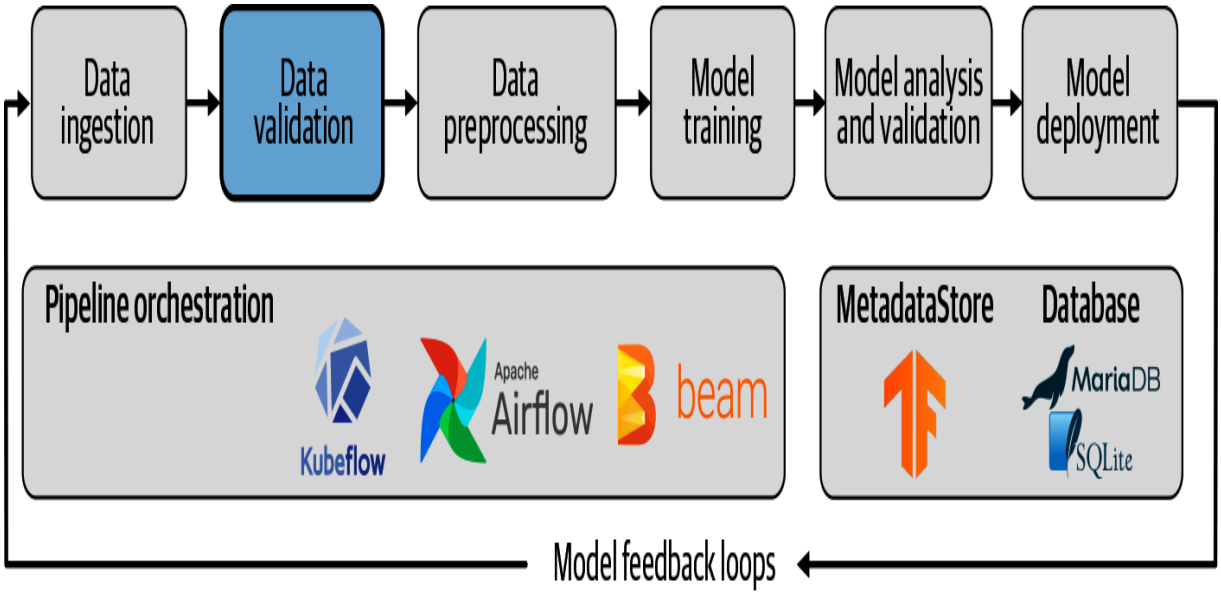


Pipeline

Machine Learning

A machine learning pipeline is a way to control and automate the workflow it takes to produce a machine learning model. Machine learning pipelines consist of multiple sequential steps that do everything from data extraction and preprocessing to model training and deployment.

Machine learning pipelines are iterative as every step is repeated to continuously improve the accuracy of the model and achieve the end goal.



An example of ML Pipeline O'Reilly

The term **Pipeline** is used generally to describe the independent sequence of steps that are arranged together to achieve a task. This task could be machine learning or not. Machine Learning Pipelines are very common but that is not the only type of pipeline that exists. Data Orchestration Pipelines are another example.

According to **Microsoft docs**, there are three scenarios:

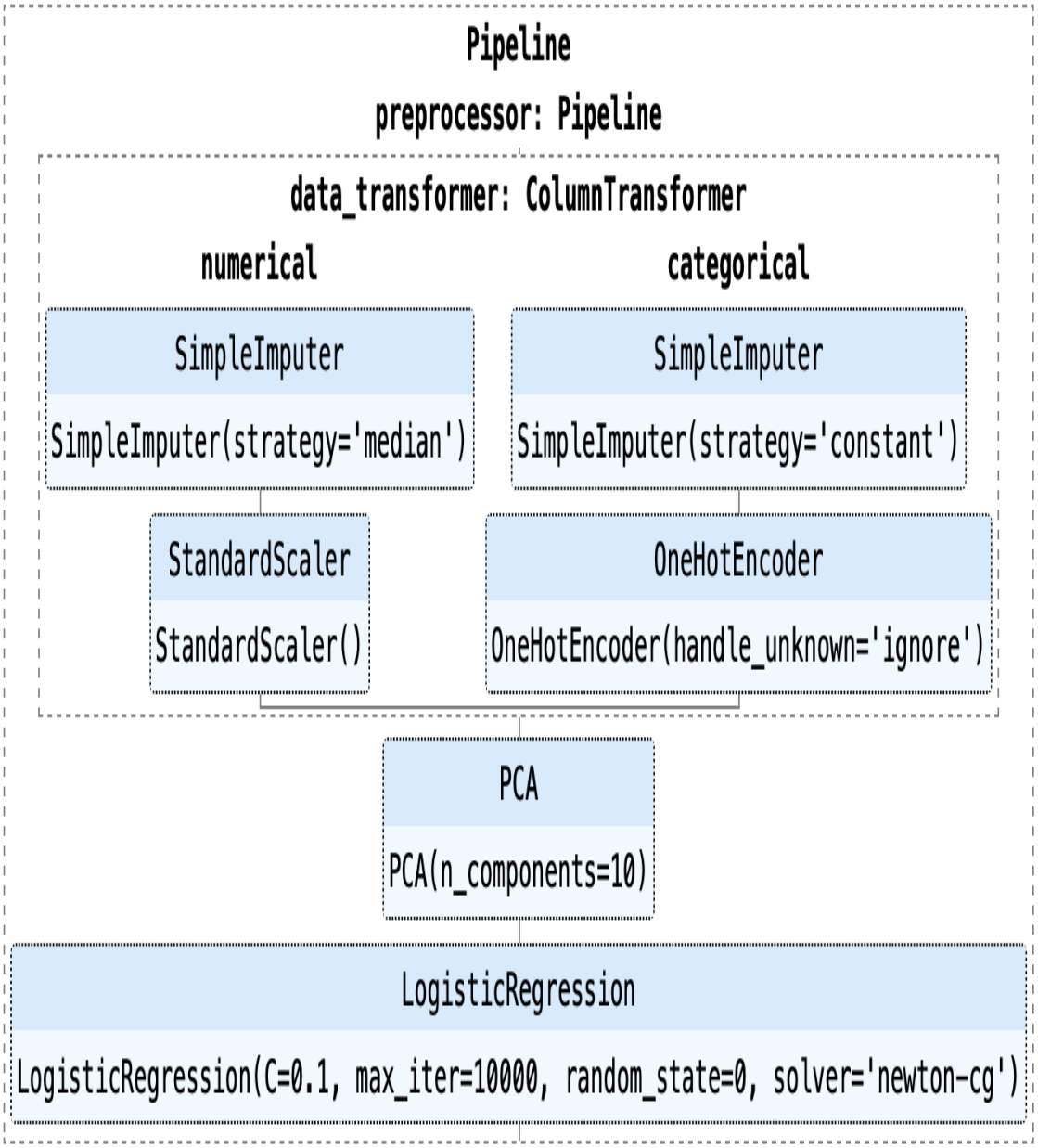
Scenario	Primary persona	Azure offering	OSS offering	Canonical pipe	Strengths
Model orchestration (Machine learning)	Data scientist	Azure Machine Learning Pipelines	Kubeflow Pipelines	Data -> Model	Distribution, caching, code-first, reuse
Data orchestration (Data prep)	Data engineer	Azure Data Factory pipelines	Apache Airflow	Data -> Data	Strongly typed movement, data-centric activities
Code & app orchestration (CI/CD)	App Developer / Ops	Azure Pipelines	Jenkins	Code + Model -> App/Service	Most open and flexible activity support, approval queues, phases with gating

Deployment

The deployment of machine learning models (or pipelines) is the process of making models available in production where web applications, enterprise software (ERPs) and APIs can consume the trained model by providing new data points, and get the predictions.

In short, Deployment in Machine Learning is the method by which you integrate a machine learning model into an existing production environment to make practical business decisions based on data. It is the last stage in the machine learning lifecycle.

Normally the term Machine Learning Model Deployment is used to describe deployment of the entire Machine Learning Pipeline, in which the model itself is only one component of the Pipeline.

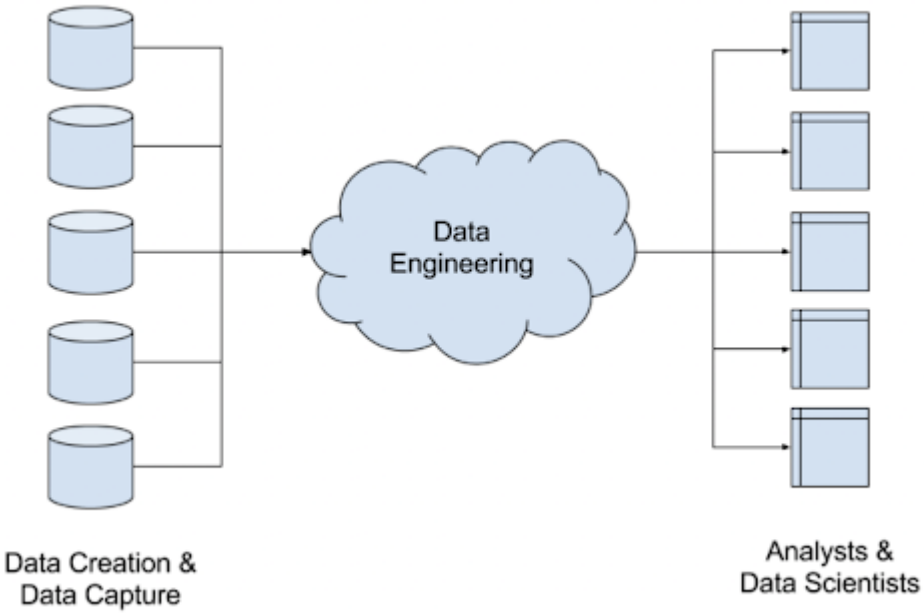


An example of a machine learning pipeline built using sklearn

As you can see in the above example, this pipeline consists of a Logistic Regression model. There are several steps in the pipeline that have to be executed first before training can begin, such as Imputation of missing values, One-Hot encoding, Scaling, and Principal Component Analysis (PCA).

Data engineering

Data engineering is the process of designing and building systems that let people collect and analyze raw data from multiple sources and formats. These systems empower people to find practical applications of the data, which businesses can use to thrive.



What Do Data Engineers Do?

Data engineering is a skill that is in increasing demand. Data engineers are the people who design the system that unifies data and can help you navigate it. Data engineers perform many different tasks including:

Acquisition: Finding all the different data sets around the business

Cleansing: Finding and cleaning any errors in the data

Conversion: Giving all the data a common format

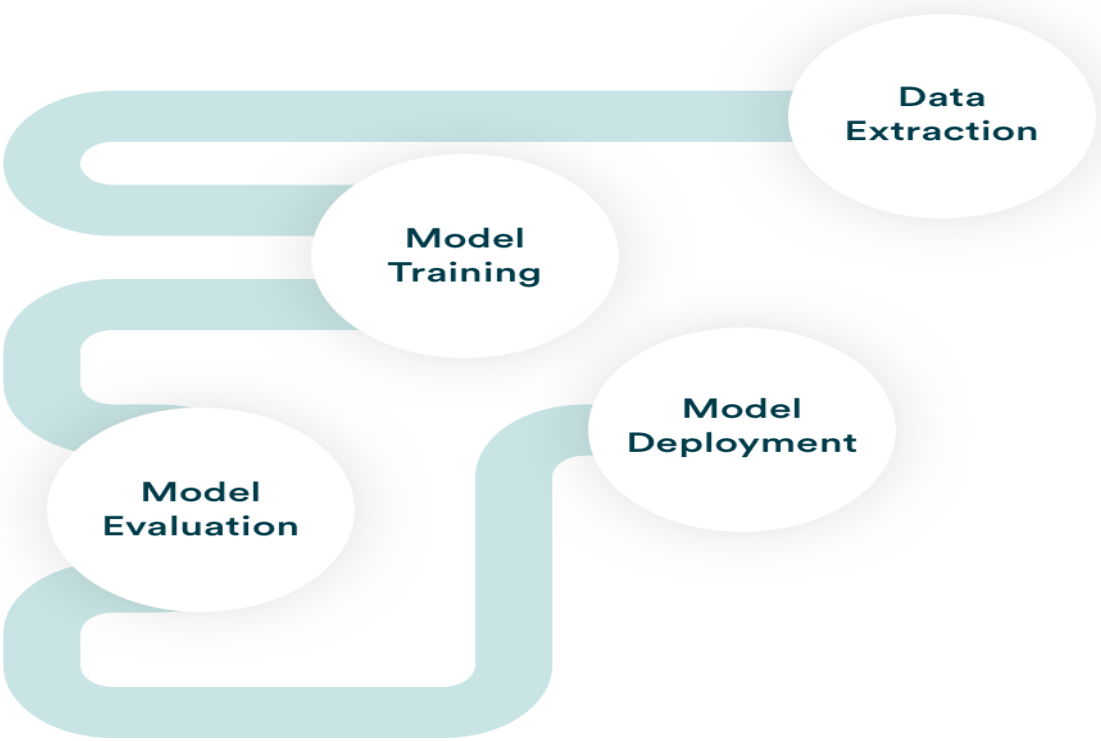
Disambiguation: Interpreting data that could be interpreted in multiple ways

Deduplication: Removing duplicate copies of data

Once this is done, data may be stored in a central repository such as a data lake or data lakehouse. Data engineers may also copy and move subsets of data into a data warehouse.

Machine Learning pipeline

A machine learning pipeline is a way to codify and automate the workflow it takes to produce a machine learning model. Machine learning pipelines consist of multiple sequential steps that do everything from data extraction and preprocessing to model training and deployment.



What is Data Science?

Data science is **the domain of study that deals with vast volumes of data using modern tools and techniques to find unseen patterns, derive meaningful information, and make business decisions**. For example, finance companies can use a customer's banking and bill-paying history to assess creditworthiness and loan risk.

How Data Science works ?

Data science uses techniques such as **machine learning and artificial intelligence to extract meaningful information and to predict future patterns and behaviours**. Advances in technology, the internet, social media, and the use of technology have all increased access to big data.

Data Science uses

data science is used in **marketing, finance, and human resources, healthcare, government programmes, and any other industry that generates data**. Marketing departments use data science to determine which product is most likely to sell.

What is Cloud?

The term Cloud refers to a Network or Internet. In other words, we can say that Cloud is something, which is present at remote location. Cloud can provide services over public and private networks, i.e., WAN, LAN or VPN.

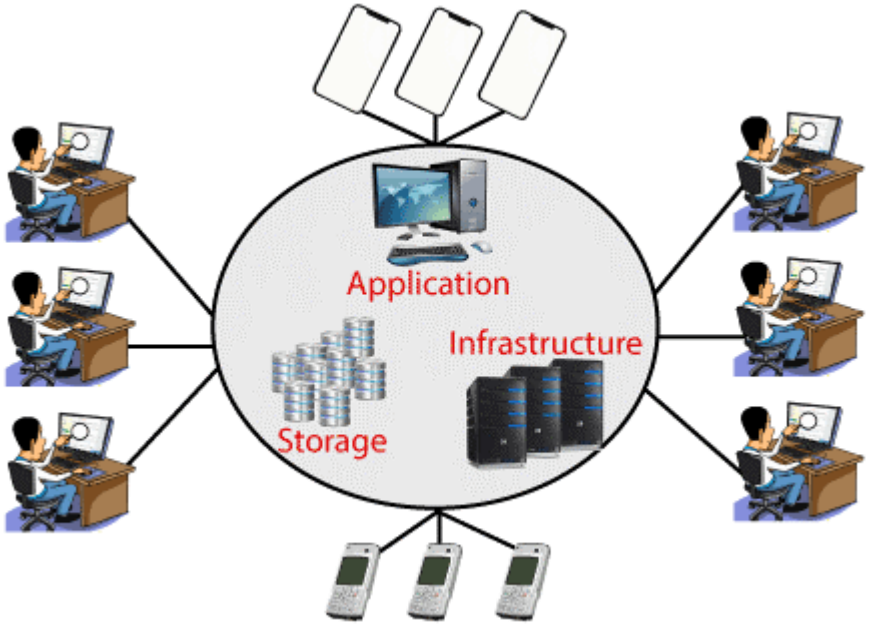
Applications such as e-mail, web conferencing, customer relationship management (CRM) execute on cloud.

Introduction to Cloud Computing

Cloud Computing refers to manipulating, configuring, and accessing the hardware and software resources remotely. It offers online data storage, infrastructure, and application.

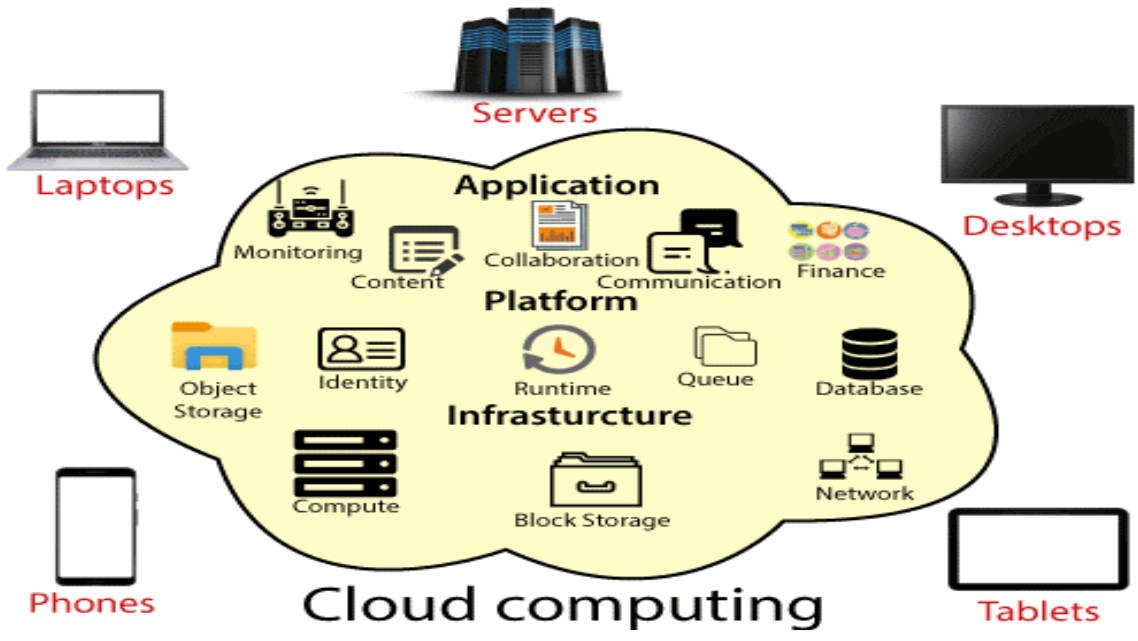
Cloud computing offers platform independency, as the software is not required to be installed locally on the PC. Hence, the Cloud Computing is making our business applications mobile and collaborative.

Cloud Computing is the delivery of computing services such as servers, storage, databases, networking, software, analytics, intelligence, and more, over the Cloud (Internet). The introduction of automation in a deployment pipeline allowed development teams to focus more on innovating and improving the end product for the user. By reducing the need for any manual tasks, teams are able to deploy new code updates much quicker and with less risk of any human error.



Cloud Computing provides an alternative to the on-premises datacentre. With an on-premises datacentre, we have to manage everything, such as purchasing and installing hardware, virtualization, installing the operating system, and any other required applications, setting up the network, configuring the firewall, and setting up storage for data. After doing all the set-up, we become responsible for maintaining it through its entire lifecycle.

But if we choose Cloud Computing, a cloud vendor is responsible for the hardware purchase and maintenance. They also provide a wide variety of software and platform as a service. We can take any required services on rent. The cloud computing services will be charged based on usage.



Advantages of cloud computing

- **Cost:** It reduces the huge capital costs of buying hardware and software.
- **Speed:** Resources can be accessed in minutes, typically within a few clicks.
- **Scalability:** We can increase or decrease the requirement of resources according to the business requirements.
- **Productivity:** While using cloud computing, we put less operational effort. We do not need to apply patching, as well as no need to maintain hardware and software. So, in this way, the IT team can be more productive and focus on achieving business goals.
- **Reliability:** Backup and recovery of data are less expensive and very fast for business continuity.
- **Security:** Many cloud vendors offer a broad set of policies, technologies, and controls that strengthen our data security.

Benefits

- ✓ Cloud Computing has numerous advantages. Some of them are listed below -
- ✓ One can access applications as utilities, over the Internet.
- ✓ One can manipulate and configure the applications online at any time.
- ✓ It does not require to install a software to access or manipulate cloud application.
- ✓ Cloud Computing offers online development and deployment tools, programming runtime environment through **PaaS model.**
- ✓ Cloud resources are available over the network in a manner that provide platform independent access to any type of clients.
- ✓ Cloud Computing offers **on-demand self-service.** The resources can be used without interaction with cloud service provider.
- ✓ Cloud Computing is highly cost effective because it operates at high efficiency with optimum utilization. It just requires an Internet connection
- ✓ Cloud Computing offers load balancing that makes it more reliable.

Essentials of Cloud Computing

Essential elements lead to flexibility



Elasticity



Faster and/or self-provisioning and de-provisioning



Application Programming Interfaces

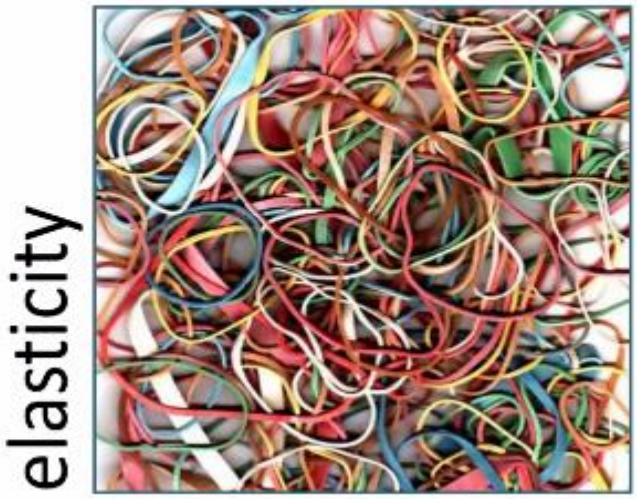


Pay-as-you-go billing model

Essential element: Elasticity

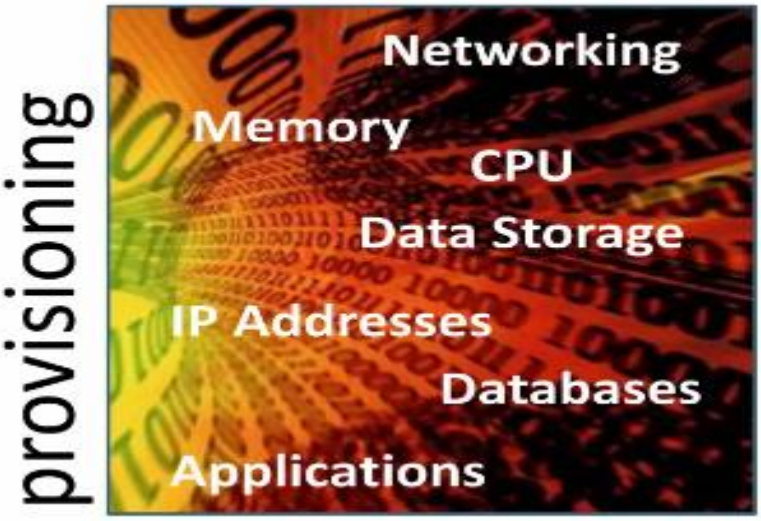
- Creating, launching, and terminating computing resources, as needed
- The more resources active and available, the more elastic

➢ Often enabled by virtualization technology



Essential element: Provisioning

- Companies or their service providers control the deployment of cloud services through a defined service catalog



Essential element: Application Program Interfaces

- Control APIs: allow cloud infrastructure to be added, reconfigured, or removed in real time
- Data APIs: are the conduits through which data flows in and out of the cloud
- Application functionality APIs: enable the functionality with which end users interact with infrastructure and data

➢ APIs will evolve as cloud offerings become more complex.



Essential element:
Billing

- Cloud computing employs a usage billing model
- Billing models include:
  - per user
  - per gigabyte (GB)
  - per server
  - pay per-use

➢ Billing types will evolve based on the imagination of the service provider and the user.

billing



Oracle cloud computing: 5 essentials

1. Rapid Elasticity
2. On-Demand self-service provisioning
3. Broad network access
4. Measure service (Billing and metering of service usage)
5. Resource pooling

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Types of Cloud Computing /Cloud Deployment Models

Different types of cloud computing deployment models are:

1. Public cloud
2. Private cloud
3. Hybrid cloud
4. Community cloud

1. Public Cloud

The public cloud makes it possible for anybody to access systems and services. The public cloud may be less secure as it is open for everyone. The public cloud is one in which cloud infrastructure services are provided over the internet to the general people or major industry groups. The infrastructure in this cloud model is owned by the entity that delivers the cloud services, not by the consumer. It is a type of cloud hosting that allows customers and users to easily access systems and services. This form of cloud computing is an excellent example of cloud hosting, in which service providers supply services to a variety of customers. In this arrangement, storage backup and retrieval services are given for free, as a subscription, or on a per-use basis. Example: Google App Engine etc.

Advantages of the public cloud model:

- **Minimal Investment:** Because it is a pay-per-use service, there is no substantial upfront fee, making it excellent for enterprises that require immediate access to resources.
- **No setup cost:** The entire infrastructure is fully subsidized by the cloud service providers, thus there is no need to set up any hardware.
- **Infrastructure Management is not required:** Using the public cloud does not necessitate infrastructure management.
- **No maintenance:** The maintenance work is done by the service provider (Not users).
- **Dynamic Scalability:** To fulfill your company's needs, on-demand resources are accessible.

2. Private Cloud

The private cloud deployment model is the exact opposite of the public cloud deployment model. It's a one-on-one environment for a single user (customer). There is no need to share your hardware with anyone else. The distinction between private and public cloud is in how you handle all of the hardware. It is also called the "internal cloud" & it refers to the ability to access systems and services within a given border or organization.

The cloud platform is implemented in a cloud-based secure environment that is protected by powerful firewalls and under the supervision of an organization's IT department. The private cloud gives the greater flexibility of control over cloud resources.

Advantages of the private cloud model:

- **Better Control:** You are the sole owner of the property. You gain complete command over service integration, IT operations, policies, and user behavior.
- **Data Security and Privacy:** It's suitable for storing corporate information to which only authorized staff have access. By segmenting resources within the same infrastructure, improved access and security can be achieved.
- **Supports Legacy Systems:** This approach is designed to work with legacy systems that are unable to access the public cloud.
- **Customization:** Unlike a public cloud deployment, a private cloud allows a company to tailor its solution to meet its specific needs.

3. Hybrid cloud

By bridging the public and private worlds with a layer of proprietary software, hybrid cloud computing gives the best of both worlds. With a hybrid solution, you may host the app in a safe environment while taking advantage of the public cloud's cost savings. Organizations can move data and applications between different clouds using a combination of two or more cloud deployment methods, depending on their needs.

Advantages of the hybrid cloud model:

- **Flexibility and control:** Businesses with more flexibility can design personalized solutions that meet their particular needs.
- **Cost:** Because public clouds provide for scalability, you'll only be responsible for paying for the extra capacity if you require it.
- **Security:** Because data is properly separated, the chances of data theft by attackers are considerably reduced.

4. Community cloud

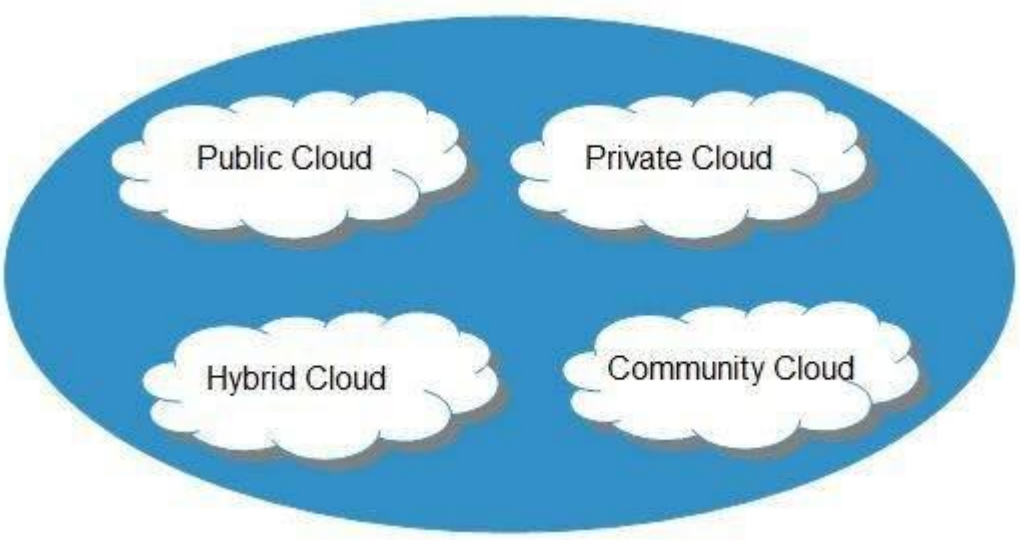
It allows systems and services to be accessible by a group of organizations. It is a distributed system that is created by integrating the services of different clouds to address the specific needs of a community, industry, or business. The infrastructure of the community could be shared between the organization which has shared concerns or tasks. It is generally managed by a third party or by the combination of one or more organizations in the community.

Advantages of the community cloud model:

- **Cost Effective:** It is cost-effective because the cloud is shared by multiple organizations or communities.
- **Security:** Community cloud provides better security.
- **Shared resources:** It allows you to share resources, infrastructure, etc. with multiple organizations.
- **Collaboration and data sharing:** It is suitable for both collaboration and data sharing.

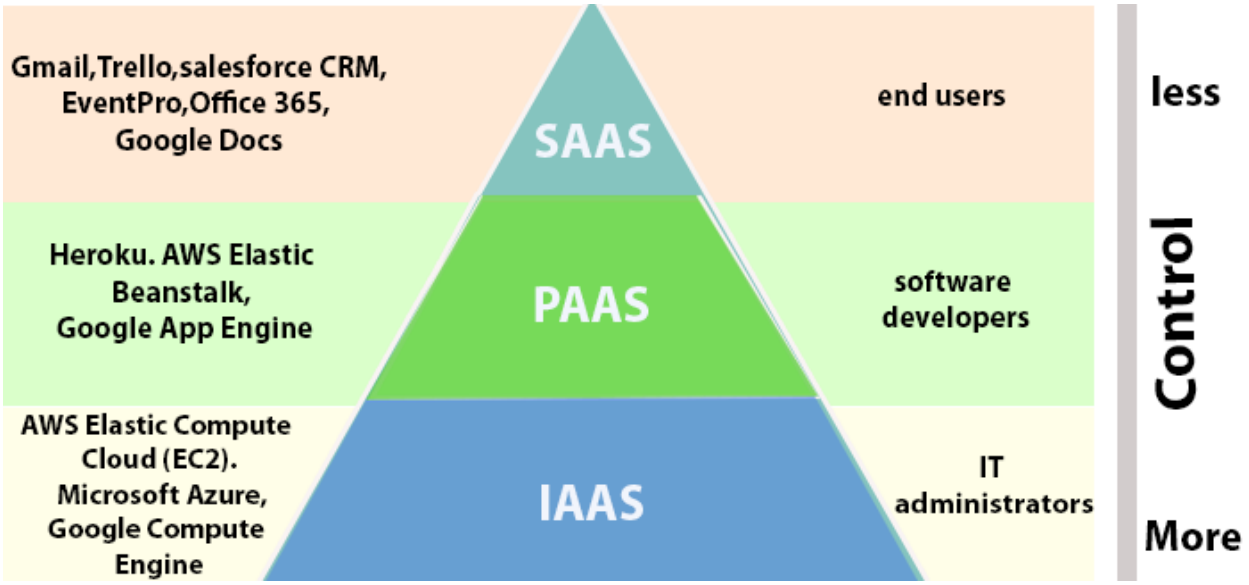
Sample Questions

Question 1. List the disadvantages of the public cloud model?



- **Public Cloud:** The cloud resources that are owned and operated by a third-party cloud service provider are termed as public clouds. It delivers computing resources such as servers, software, and storage over the internet
- **Private Cloud:** The cloud computing resources that are exclusively used inside a single business or organization are termed as a private cloud. A private cloud may physically be located on the company's on-site datacentre or hosted by a third-party service provider.
- **Hybrid Cloud:** It is the combination of public and private clouds, which is bounded together by technology that allows data applications to be shared between them. Hybrid cloud provides flexibility and more deployment options to the business.
- **Community Cloud :** The community cloud allows systems and services to be accessible by a group of organizations

Cloud Service Models / Types of Cloud Services



1. **Infrastructure as a Service (IaaS):** In IaaS, we can rent IT infrastructures like servers and virtual machines (VMs), storage, networks, operating systems from a cloud service vendor. We can create VM running Windows or Linux and install anything we want on it. Using IaaS, we don't need to care about the hardware or virtualization software, but other than that, we do have to manage everything else. Using IaaS, we get maximum flexibility, but still, we need to put more effort into maintenance.
2. **Platform as a Service (PaaS):** This service provides an on-demand environment for developing, testing, delivering, and managing software applications. The developer is responsible for the application, and the PaaS vendor provides the ability to deploy and run it. Using PaaS, the flexibility gets reduce, but the management of the environment is taken care of by the cloud vendors.
3. **Software as a Service (SaaS):** It provides a centrally hosted and managed software services to the end-users. It delivers software over the internet, on-demand, and typically on a subscription basis. E.g., Microsoft One Drive, Dropbox, WordPress, Office 365, and Amazon Kindle. SaaS is used to minimize the operational cost to the maximum extent.

Difference between IaaS, PaaS, and SaaS

The below table shows the difference between IaaS, PaaS, and SaaS -

IaaS	Paas	SaaS
It provides a virtual data center to store information and create platforms for app development, testing, and deployment.	It provides virtual platforms and tools to create, test, and deploy apps.	It provides web software and apps to complete business tasks.
It provides access to resources such as virtual machines, virtual storage, etc.	It provides runtime environments and deployment tools for applications.	It provides software as a service to the end-users.
It is used by network architects.	It is used by developers.	It is used by end users.
IaaS provides only Infrastructure.	PaaS provides Infrastructure+Platform.	SaaS provides Infrastructure+Platform

Serverless Services

Serverless computing is a method of providing backend services on an as-used basis. A serverless provider allows users to write and deploy code without the hassle of worrying about the underlying infrastructure. A company that gets backend services from a serverless vendor is charged based on their computation and do not have to reserve and pay for a fixed amount of bandwidth or number of servers, as the service is auto-scaling.

Serverless computing allows developers to purchase backend services on a flexible 'pay-as-you-go' basis, meaning that developers only have to pay for the services they use. This is like switching from a cell phone data plan with a monthly fixed limit, to one that only charges for each byte of data that actually gets used.

Serverless is a cloud-native development model that allows developers to build and run applications without having to manage servers. There are still servers in serverless, but they are abstracted away from app development

What are the advantages of serverless computing?

- ✓ **Lower costs** - Serverless computing is generally very cost-effective, as traditional cloud providers of backend services (server allocation) often result in the user paying for unused space or idle CPU time.

- ✓ **Simplified scalability** - Developers using serverless architecture don't have to worry about policies to scale up their code. The serverless vendor handles all of the scaling on demand.

- ✓ **Simplified backend code** - With FaaS, developers can create simple functions that independently perform a single purpose, like making an API call.

    - **Quicker turnaround** - Serverless architecture can significantly cut time to market. Instead of needing a complicated deploy process to roll out bug fixes and new features, developers can add and modify code on a piecemeal basis.

Major Cloud Service Providers

The top 10 cloud service providers globally in 2022 are ranked in the following table, which includes the number of regions and availability zones that each vendor possesses:

#	Cloud Service Provider	Regions	Availability Zones
1	Amazon Web Services (AWS)	26	84
2	Microsoft Azure	60	116
3	Google Cloud Platform (GCP)	34	103
4	Alibaba Cloud	27	84
5	Oracle Cloud	38	46
6	IBM Cloud (Kyndryl)	11	29
7	Tencent Cloud	21	65
8	OVHcloud	13	33
9	DigitalOcean	8	14
10	Linode (Akamai)	11	11



Virtualization in Cloud Computing

Virtualization is the "creation of a virtual (rather than actual) version of something, such as a server, a desktop, a storage device, an operating system or network resources".

In other words, Virtualization is a technique, which allows to share a single physical instance of a resource or an application among multiple customers and organizations. It does by assigning a logical name to a physical storage and providing a pointer to that physical resource when demanded.

Ex : VM WARE, Dockerc, virtual Box

What is the concept behind the Virtualization?

Creation of a virtual machine over existing operating system and hardware is known as Hardware Virtualization. A Virtual machine provides an environment that is logically separated from the underlying hardware.

The machine on which the virtual machine is going to create is known as **Host Machine** and that virtual machine is referred as a **Guest Machine.**

Types of Virtualization:

1. Hardware Virtualization.
2. Operating system Virtualization.

3. Server Virtualization.
4. Storage Virtualization.

Amazon Web Services (AWS)

Amazon Augmented AI (Amazon A2I)
Amazon CodeGuru
Amazon CodeWhisperer
Amazon DevOps Guru
Amazon Elastic Inference
Amazon Forecast
Amazon Fraud Detector
Amazon Kendra
Amazon Elastic Container Service (Amazon ECS)
Amazon Comprehend
Amazon Monitron
Amazon Lex
Amazon Lookout for Equipment
Amazon Lookout for Metrics
Amazon Lookout for Vision
Amazon Monitron
Amazon Personalize
Amazon Polly
AWS Cost and Usage Reports
AWS Inferentia
AWS Panorama
Apache MXNet on AWS
AWS Deep Learning Containers
AWS deepracer
AWS deepcomposer
AWS Inferentia
Computer vision
Analyse images and videos
Detect defects and automate inspection
Utilize computer vision at the edge
Extract text and data
Acquire insights

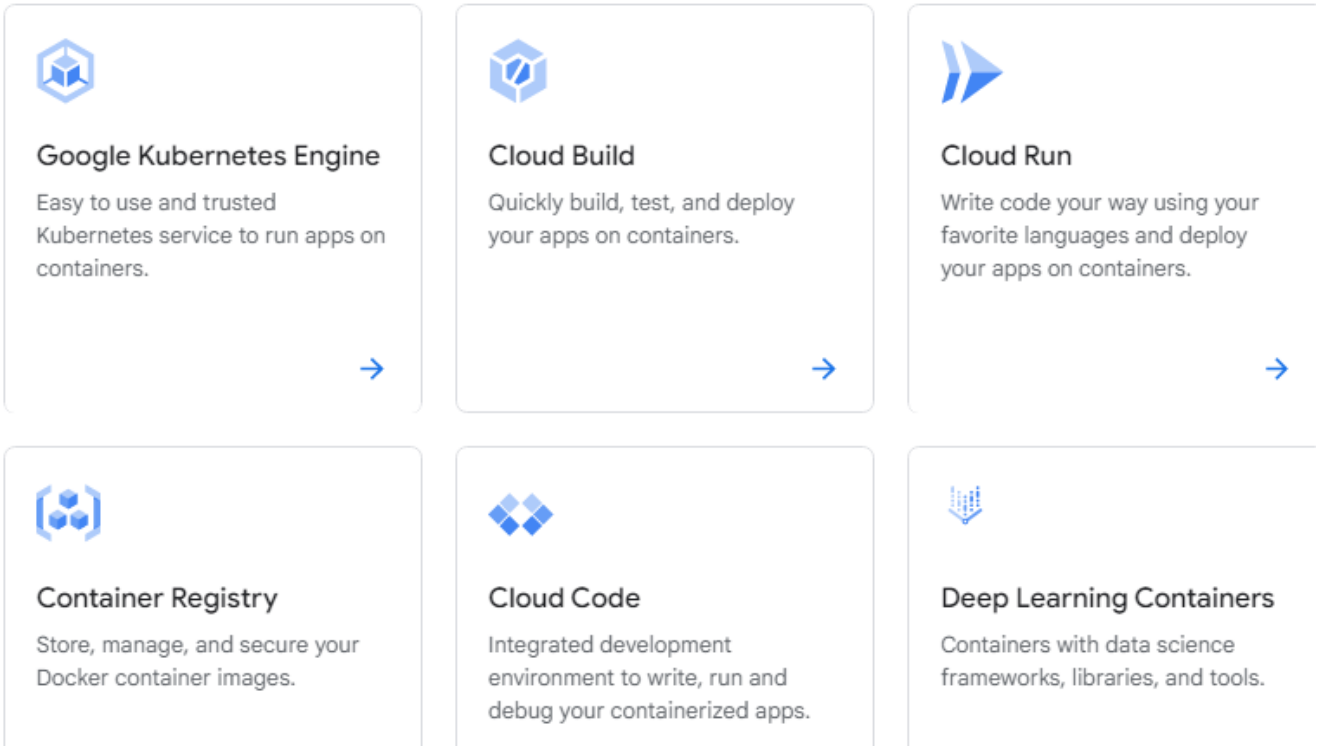
What are Containers?

Containers are packages of software that contain all of the necessary elements to run in any environment. In this way, containers virtualize the operating system and run anywhere, from a private data center to the public cloud or even on a developer's personal laptop. From Gmail to YouTube to Search, everything at Google runs in containers. Containerization allows our development teams to move fast, deploy software efficiently, and operate at an unprecedented scale. We've learned a lot about running containerized workloads and we've shared this knowledge with the community along the way.

What are containers used for?

Containers are a form of operating system virtualization. A single container might be used to run anything from a small microservice or software process to a larger application. Inside a container are all the necessary executables, binary code, libraries, and configuration files.

Containers offer a logical packaging mechanism in which applications can be abstracted from the environment in which they actually run. This decoupling allows container-based applications to be deployed easily and consistently, regardless of whether the target environment is a private data center, the public cloud, or even a developer's personal laptop.



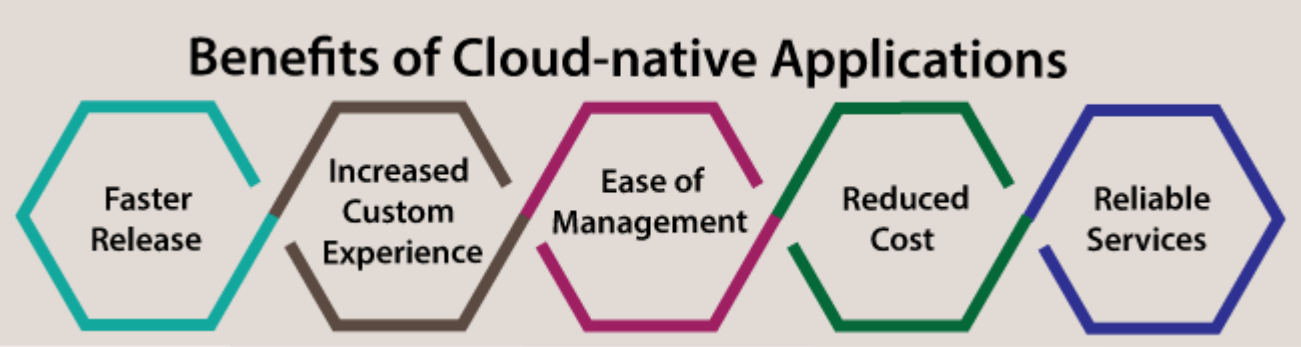
Cloud-Native Applications

The Cloud-Native applications are a group of independent, small, loosely-coupled services. Cloud-native applications are created for delivering well-identified business value, as an ability to incorporate the feedback of users for extended improvements. In other words, cloud-native applications improvement is a way for speeding-up how we create applications users wish at the pace an enterprise requires.

If an application is Cloud-native, specifically, it's created for facilitating automated management and consistent development experience across hybrid, public, and private clouds. Enterprises use cloud computing for increasing the availability and scalability of applications. These aspects are accomplished through on-demand and self-service provisioning of various resources. It is also automating the lifecycle of an application to production from development.

The development of Cloud-native applications is an approach for quickly updating and building applications while reducing risk and improving quality. More specifically, it is a way to run and build fault-tolerant, scalable, and responsive applications anywhere inside the hybrid, public, and private clouds.

Cloud-native applications



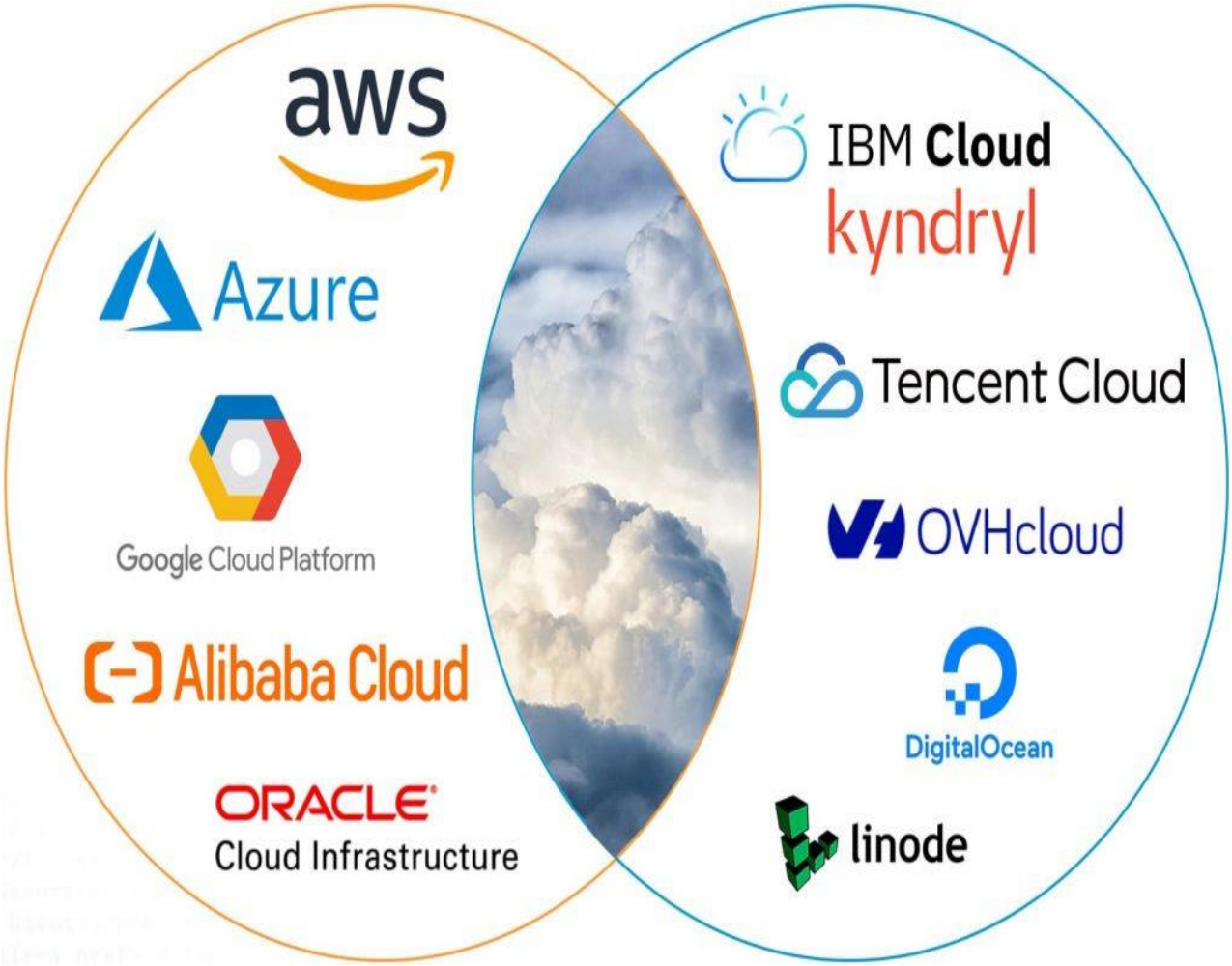
- **Faster release:** Faster deployment displays responsiveness and agility, which are essential to remain connected in today's changing business world rapidly. If we require explaining faster deployment, we require explaining DevOps because it provides collaboration and finds the way for quick delivery. The Cloud-native applications are often the outcome of any successful DevOps as operations, development. Testing groups can efficiently and easily coordinate for each release, even when they are within geographically distinct places.
- **Increased customer experience:** It is necessary to build a business model, i.e., **customer-centric**. Constant incremental developments are crucial to the applications for improving customer satisfaction. These iterations and changes can be easily carried out. Thus, it provides us a competitive edge needed to be customer-friendly and more engaging.

- **Ease of management:** Various cloud automation comes with advanced and comprehensive ecosystems that make framework management easy. Constantly, the cloud is evolving to contain even the complex application over its own.
- **Reduced cost:** The cloud provides **Containerization**. The containerization process is supported via Container-orchestration open-source systems such as **Kubernetes**. Kubernetes currently is an industry-standard to manage resources on any cloud. Containerization permits us to build Cloud-native abilities like compute time (pay-per-use) and server-less in milliseconds. It will add flexibility in the price to deploy and build applications.
- **Reliable services:** We can use the architecture of micro-services which can isolate and identify the failure impact to minimize the downtime using Cloud-native applications. With Containerization and micro-services, our applications can be made to become self-heal and fault-tolerant. Also, it improves customer-experience as the customers can like a highly reliable achievement.

Explore AI (ML and DL) services across public cloud platforms

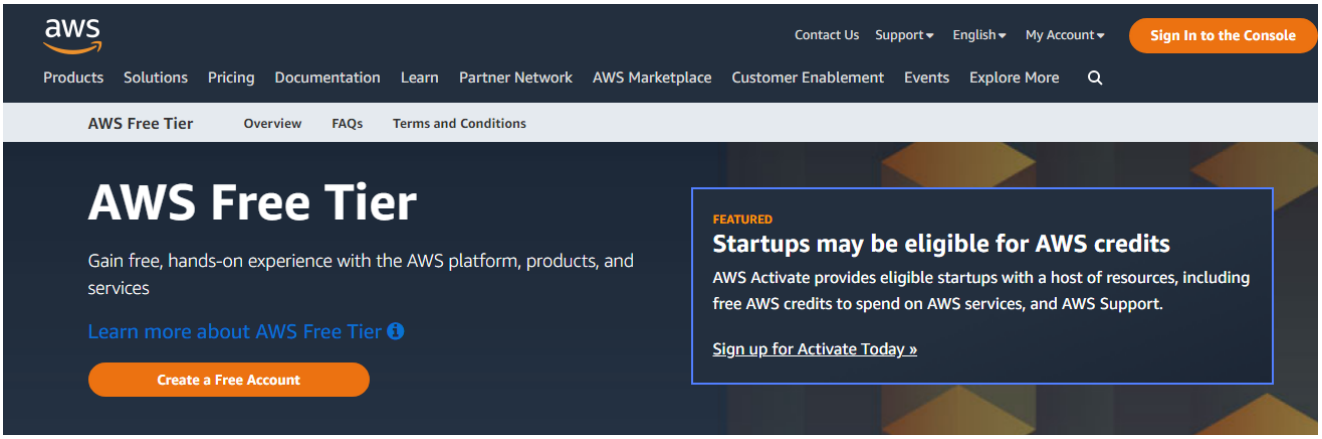
Getting to know cloud platforms



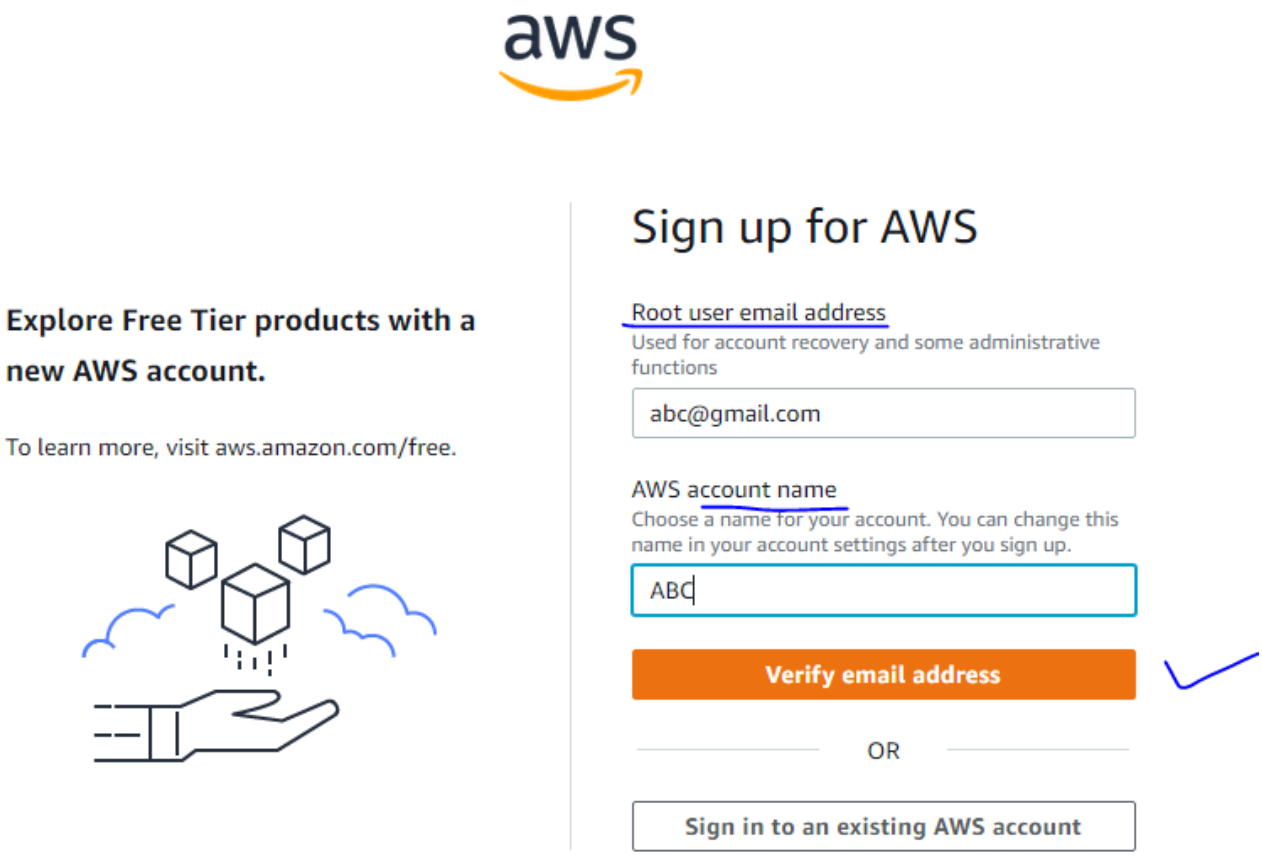


Creating an account AWS ACCOUNT

1.Select Amazon Web Series (AWS) as Free Tier Cloud Account by typing the URL as https://aws.amazon.com/account/signup



2.Click on "Create a free Account "and it will navigate to next Page



3.Enter Email ID of your choice for Root User AWS Account. It will send the Verification CODE for the Email ID which is mentioned above.

aws

Explore Free Tier products with a new AWS account.

To learn more, visit aws.amazon.com/free.



Sign up for AWS

Confirm you are you

Making sure you are secure -- it's what we do.

We sent an email with a verification code to **abc@gmail.com**. (not you?)

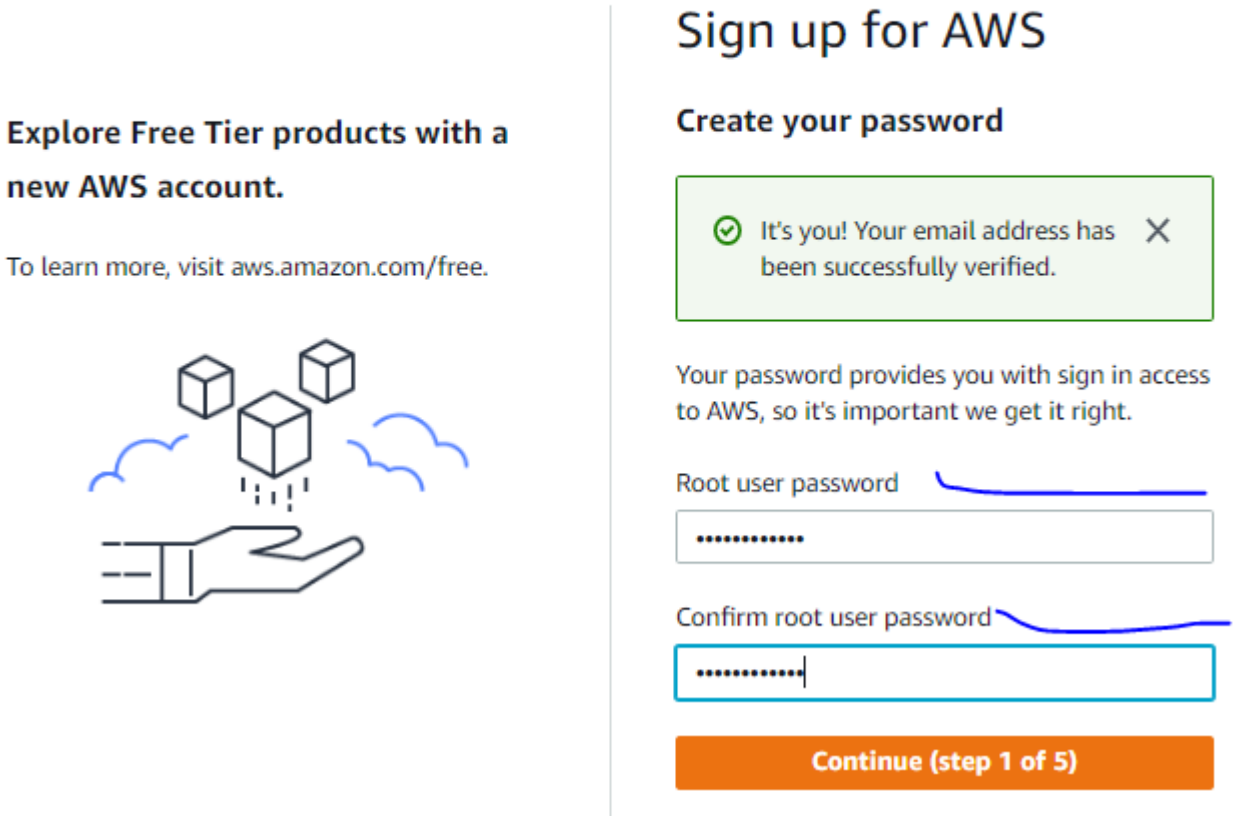
Enter it below to confirm your email.

Verification code

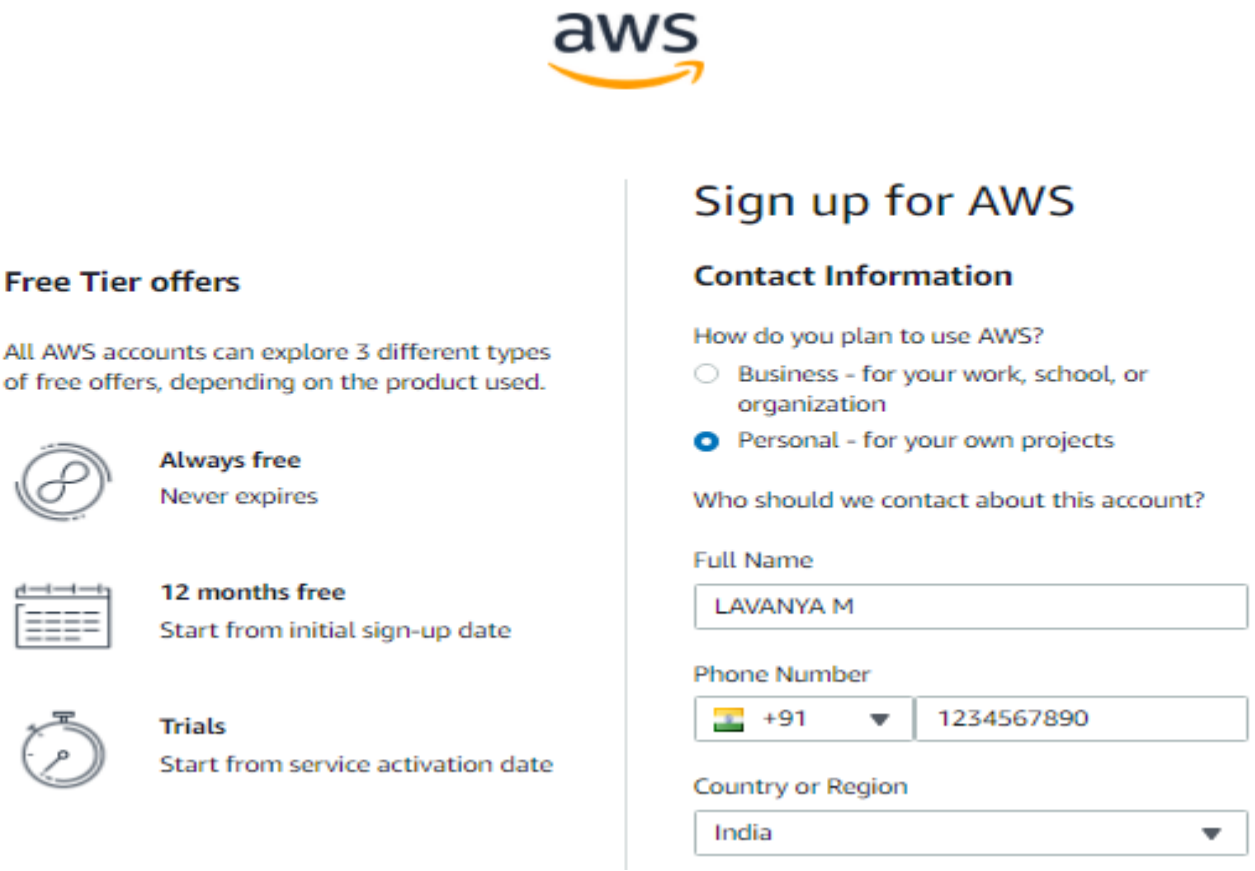
123456

Verify

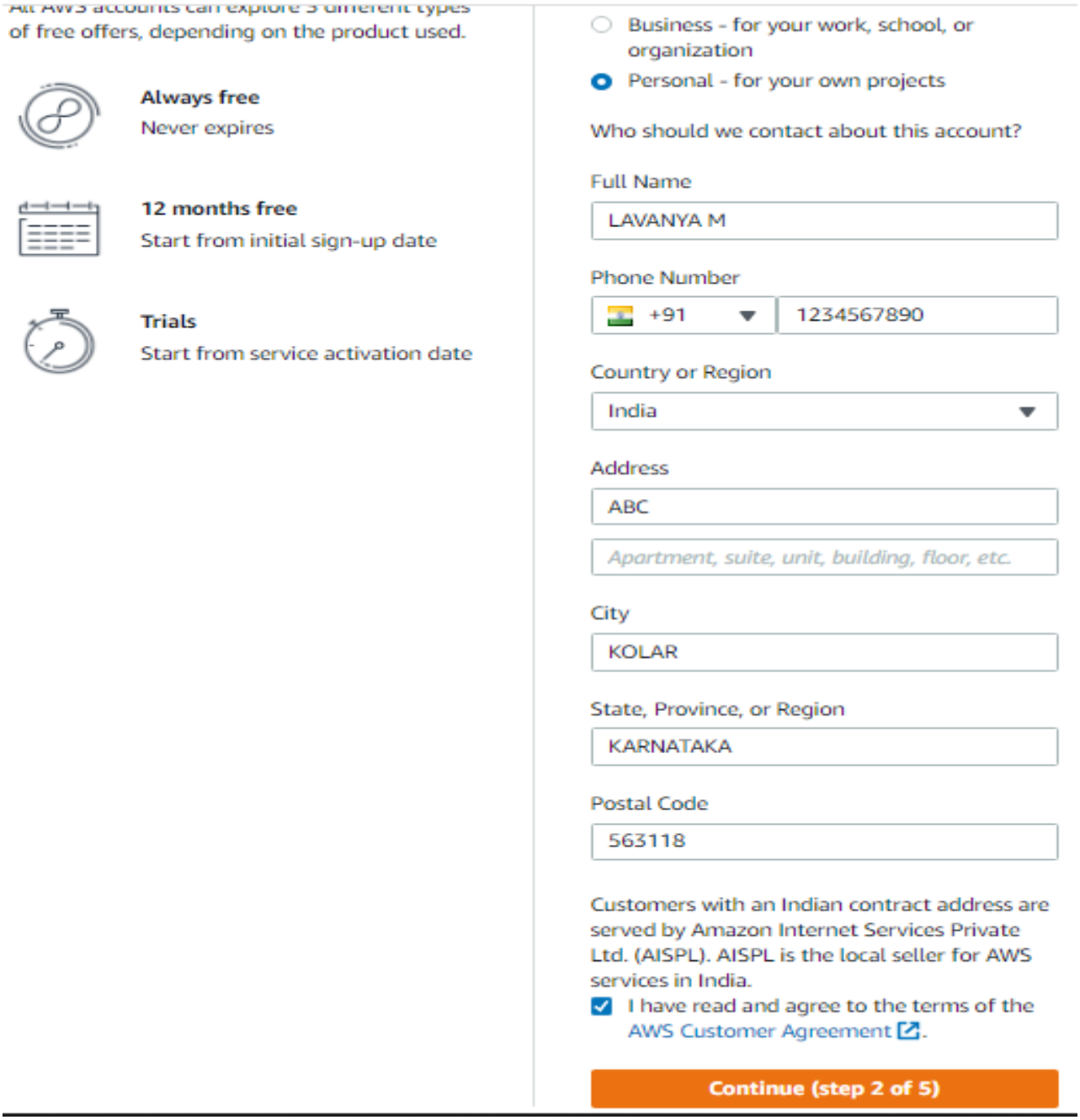
Resend code



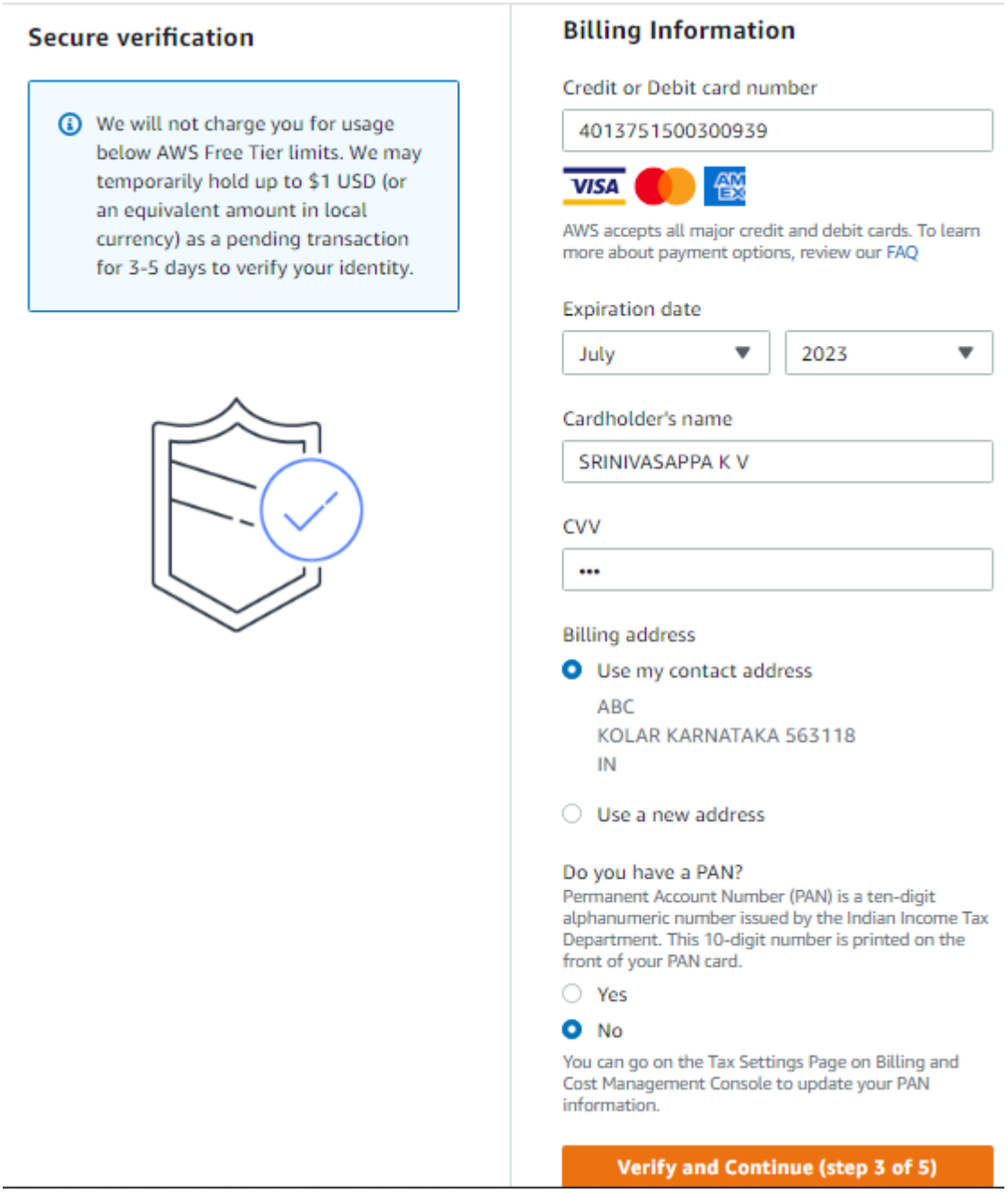
4.After Successful verification of email it will as Password for Root User AWS Account.



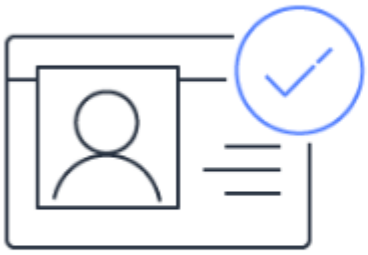
5.Enter Full Name and Phone Number select Country/region as India.

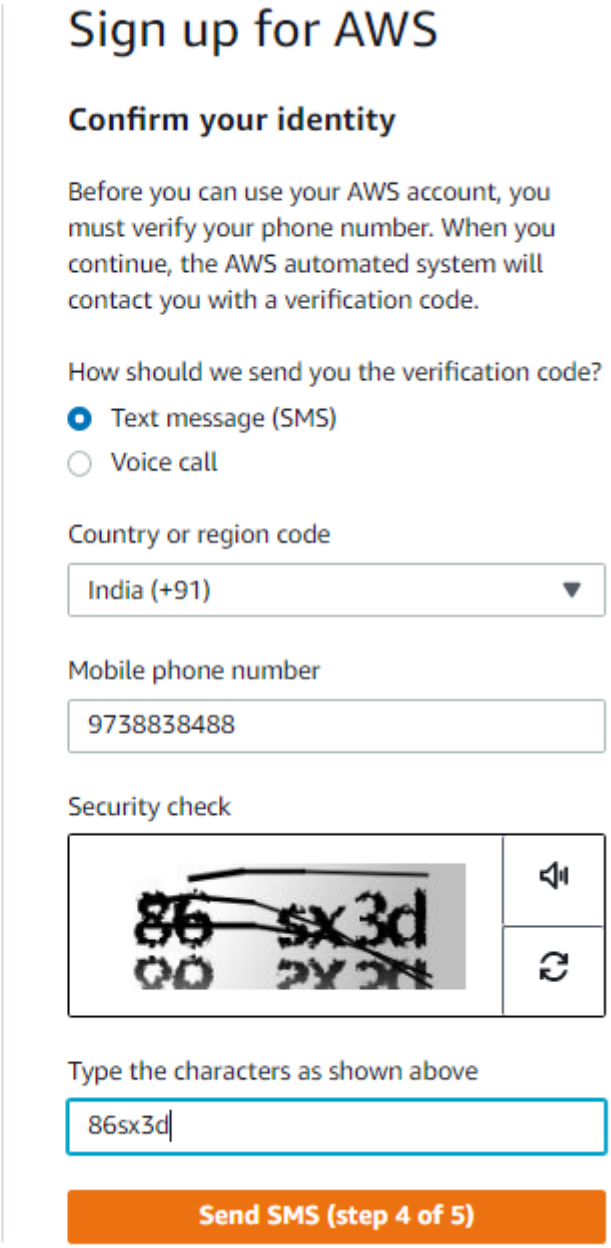


6.Enter Address as City, State and Pin Code

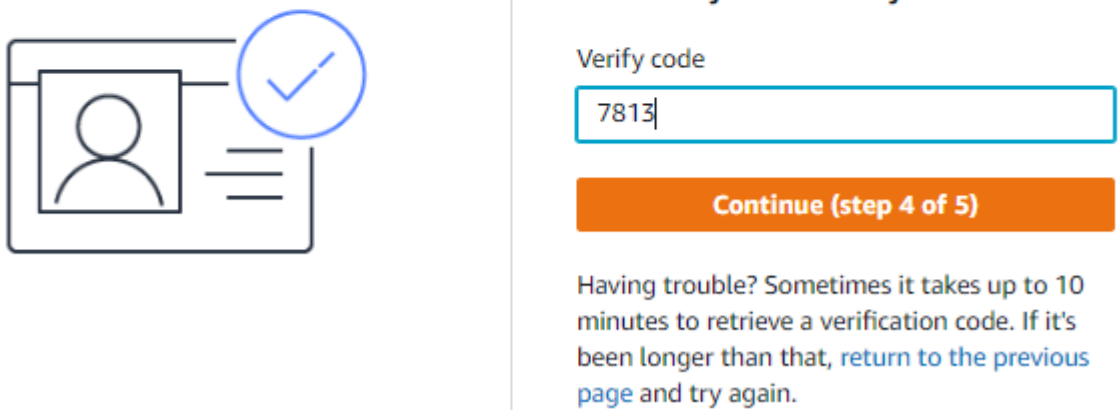


7.Enter Billing Information like Credit/Debit Card details like Number, Expiry date, Card Holder Name, CVV. It is mandatory to create Free Tier account

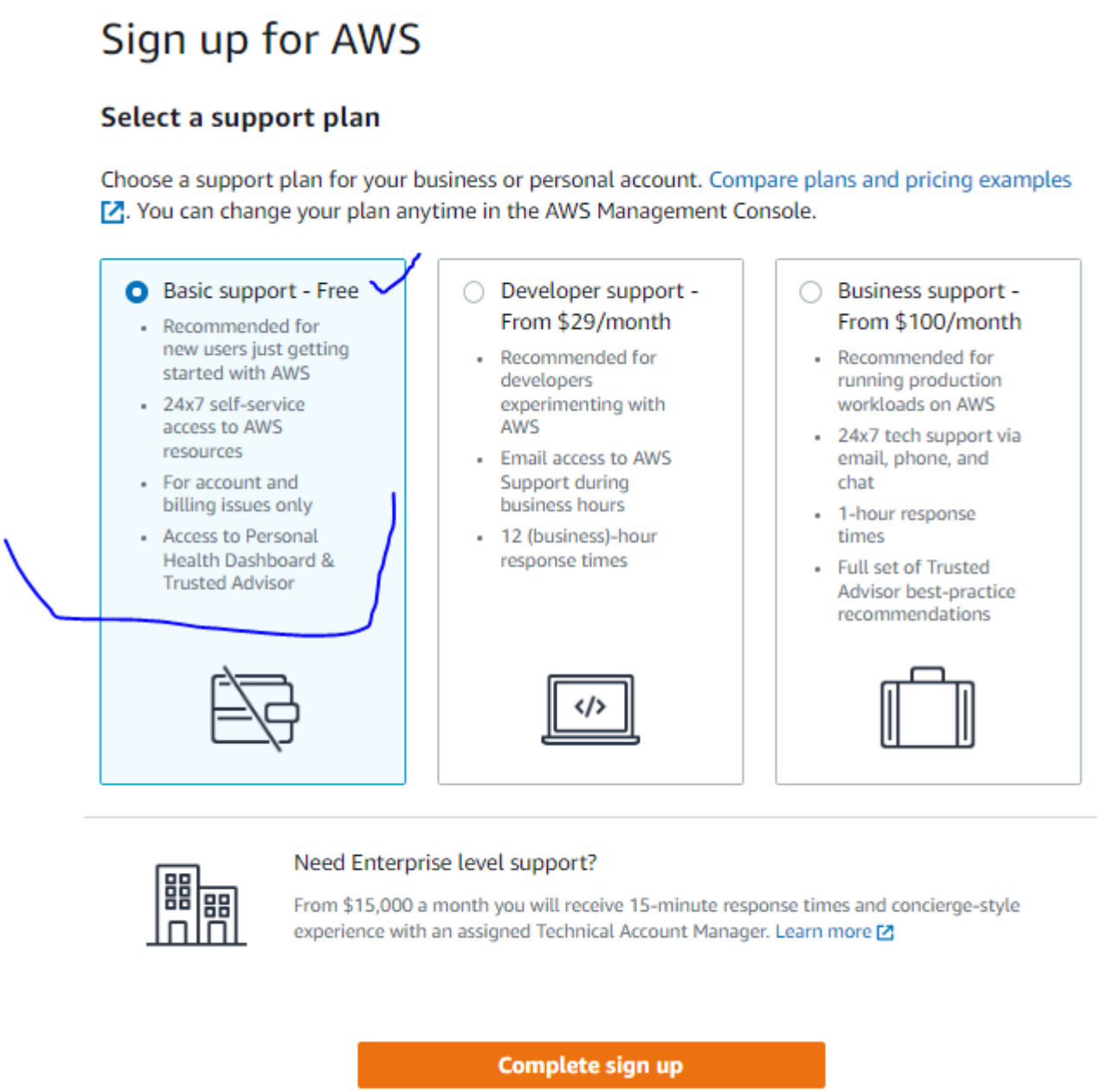




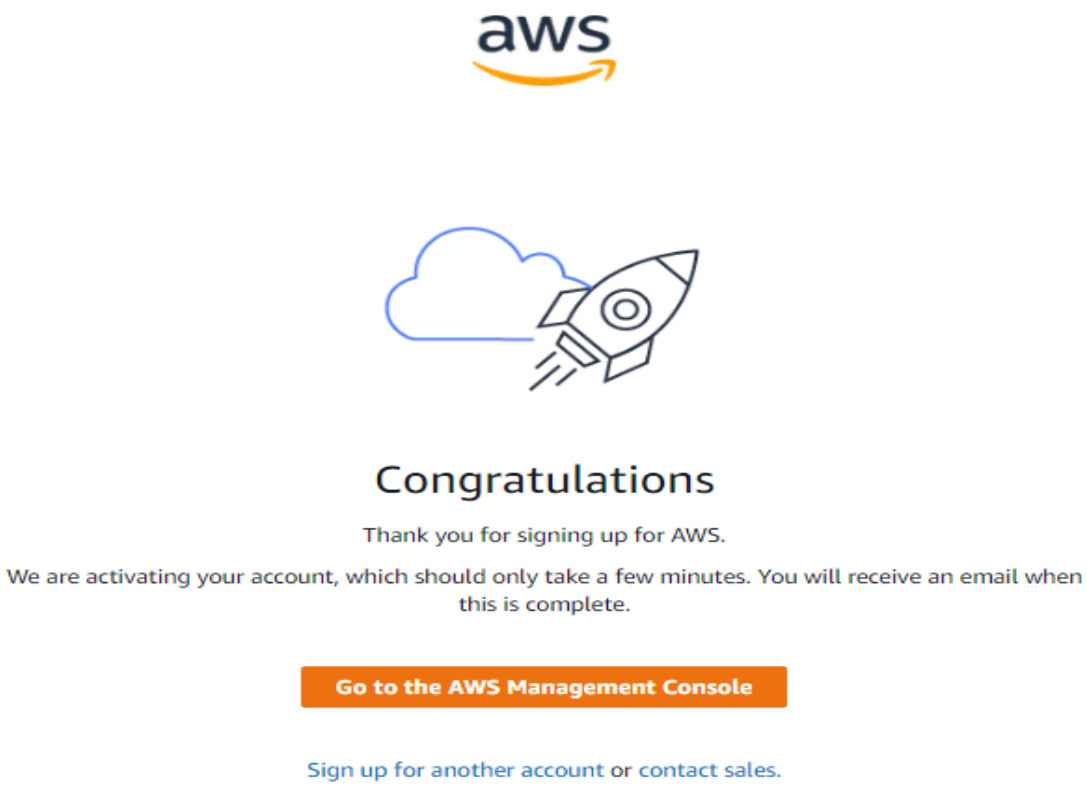
8.Confirm your Identity through SMS by giving Original Mobile number.



9.You will receive Verification Code and enter the same



10.Always select basic Support- free from the above three options



cloud SLA

A cloud SLA (cloud service-level agreement) is an agreement between a cloud service provider and a customer that ensures a minimum level of service is maintained. It guarantees levels of reliability, availability and responsiveness to systems and applications; specifies who governs when there is a service interruption; and describes penalties if service levels are not met.

A cloud infrastructure can span geographies, networks and systems that are both physical and virtual. While the exact metrics of a cloud SLA can vary by service provider, the areas covered are uniform:

- volume and quality of work (including precision and accuracy);
- speed;
- responsiveness; and
- efficiency.

The SLA document aims to establish a mutual understanding of the services, prioritized areas, responsibilities, guarantees and warranties provided by the service provider. It clearly outlines metrics and responsibilities among the parties involved in cloud configurations, such as the specific amount of response time to report or address system failures.

The importance of a cloud SLA

Service-level agreements are fundamental as more organizations rely on external providers for their critical systems, applications and data. A cloud SLA ensures cloud providers meet certain enterprise-level requirements and provide customers with a clearly defined set of deliverables. It also describes financial penalties, such as credits for service time, if the provider fails to live up to the guaranteed terms.

Basis Of	IAAS	PAAS	SAAS
Stands for	Infrastructure as a service.	Platform as a service.	Software as a service.
Uses	IAAS is used by network architects.	PAAS is used by developers.	SAAS is used by the end user.
Access	IAAS gives access to the resources like virtual machines and virtual storage.	PAAS gives access to run time environment to deployment and development tools for application.	SAAS gives access to the end user.
Model	It is a service model that provides virtualized computing resources over the internet.	It is a cloud computing model that delivers tools that are used for the development of applications.	It is a service model in cloud computing that hosts software to make it available to clients.
Technical understanding.	It requires technical knowledge.	Some knowledge is required for the basic setup.	There is no requirement about technicalities company handles everything.
Popularity.	It is popular among developers and researchers.	It is popular among developers who focus on the development of apps and scripts.	It is popular among consumers and companies, such as file sharing, email, and networking.
Cloud services.	Amazon Web Services, sun, vCloud Express.	Facebook, and Google search engine.	MS Office web, Facebook and Google Apps.
Enterprise services.	AWS virtual private cloud.	Microsoft Azure.	IBM cloud analysis.
Outsourced cloud services.	Salesforce	Force.com, Gigaspaces.	AWS, Terremark
User Controls	Operating System, Runtime, Middleware, and Application data	Data of the application	Nothing

What is Big Data?

Big Data is a collection of data that is huge in volume, yet growing exponentially with time. It is a data with so large size and complexity that none of traditional data management tools can store it or process it efficiently. Big data is also a data but with huge size.

The **New York Stock Exchange** is an example of Big Data that generates about ***one terabyte*** of new trade data per day.

Vs of Big Data

Big data is a collection of data from many different sources and is often describe by five characteristics: volume, value, variety, velocity, and veracity.

- **Volume:** the size and amounts of big data that companies manage and analyze
- **Value:** the most important "V" from the perspective of the business, the value of big data usually comes from insight discovery and pattern recognition that lead to more effective operations, stronger customer relationships and other clear and quantifiable business benefits
- **Variety:** the diversity and range of different data types, including unstructured data, semi-structured data and raw data
- **Velocity:** the speed at which companies receive, store and manage data – e.g., the specific number of social media posts or search queries received within a day, hour or other unit of time
- **Veracity:** the "truth" or accuracy of data and information assets, which often determines executive-level confidence

Sources of Big Data

A significant part of big data is generated from three primary resources:

- Machine data
- Social data, and
- Transactional data.

In addition to this, companies also generate data internally through direct customer engagement. This data is usually stored in the company's firewall. It is then imported externally into the management and analytics system.

Machine Data

Machine data is automatically generated, either as a response to a specific event or a fixed schedule. It means all the information is developed from multiple sources such as smart sensors, SIEM logs, medical devices and wearables, road cameras, IoT devices, satellites, desktops, mobile phones, industrial machinery, etc. These sources enable companies to track consumer behaviour. Data extracted from machine sources grow exponentially along with the changing external environment of the market.

Social Data

It is derived from social media platforms through tweets, retweets, likes, video uploads, and comments shared on Facebook, Instagram, Twitter, YouTube, Linked In etc. The extensive data generated through social media platforms and online channels offer qualitative and quantitative insights on each crucial facet of brand-customer interaction

Transactional Data

As the name suggests, transactional data is information gathered via online and offline transactions during different points of sale. The data includes vital details like transaction time, location, products purchased, product prices, payment methods, discounts/coupons used, and other relevant quantifiable information related to transactions.

The sources of transactional data include:

- Payment orders
- Invoices
- Storage records and
- E-receipts

Transactional data is a key source of business intelligence. The unique characteristic of transactional data is its time print. Since all transactional data include a time print, it is time-sensitive and highly volatile. In plain words, transactional data will lose its credibility and importance if not used in due time. Thus, companies using transactional data promptly can gain the upper hand in the market.

Some additional steps ideal to the analysis of big data are:

- ✓ Deep learning offshoot of data
- ✓ Data mining
- ✓ Streaming analytics
- ✓ Predictive modelling
- ✓ Statistical analysis
- ✓ Text mining

Role of Big Data in AI&ML

How can big data be used in AI and machine learning?

Machine learning systems use data-driven algorithms and statistical models to analyze and find patterns in data. This is different from traditional rules-based approaches that follow explicit instructions. Big data provides the raw material by which machine learning systems can derive insights

What is the difference between artificial intelligence and big data?

Big data refers to large volumes of diverse and dynamic data that can be mined for information. AI is a set of technologies that enables machines to simulate human intelligence. AI requires the volumes of big data to effectively learn and evolve. Big data relies on AI to more intelligently mine for information.

Why big data influence the rise artificial intelligence?

AI is developed by using machine learning which inputs data for the algorithm to learn responses. Big data is a valuable source of data for machine learning, but the extent of their potential contribution is uncertain, given that big data often involves unstructured free text

Deep learning

"Deep learning allows computational models that are composed of multiple processing layers to learn representations of data with multiple levels of abstraction".

The notion of deep learning is built on the core of classical machine learning approaches (like regression analysis, classification algorithms, the use of optimization algorithms, etc.) and proves worthy over them in the following situations:

- when enormous amounts of data are available
- when enormous computational power is available
- a non-interpretable model is not detrimental

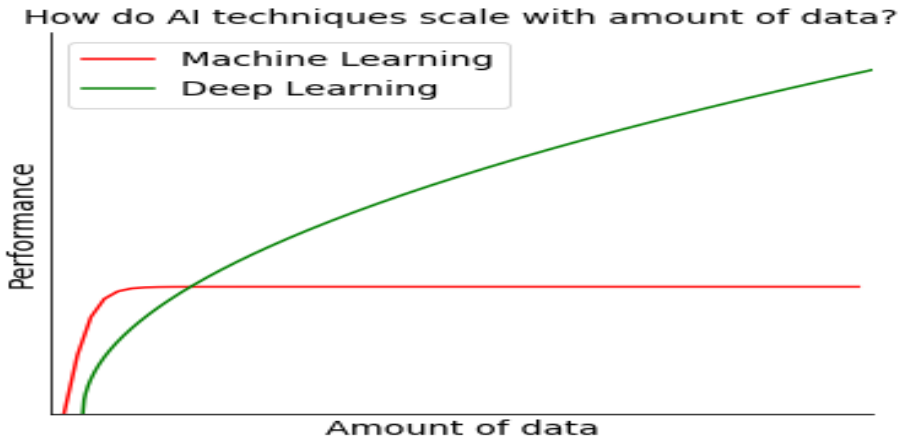
But how exactly does deep learning improve upon classical machine learning? To answer this question let us consider the cat scenario again.

If a classical machine learning algorithm has to identify a cat's face, then it would rely on feature engineering to build its model. The quality of features decides the accuracy of identification. On the contrary, a deep learning algorithm doesn't need feature engineering from the user end! Instead, it asks for a sequence of input images and creates features (whisker, eyes, nose, hair pattern, etc.) out of them, all by itself! In this sense, we can say that a deep learning algorithm learns just as a child does.

The higher the number of cats a child observes, the more accurate his/her ability to identify a cat becomes. Similarly, the more the number of training data samples a deep learning algorithm gets, the higher its accuracy becomes in identifying cats becomes.

Of course, similar to a child, a deep learning algorithm does perform mistakes while starting its learning journey but due to the availability of enormous data and computational resources in the present times it has proven to be far better than classical machine learning architectures in solving certain complicated problems (as seen before) and in some cases competitive to the human experts.

The following graph sums up the story. Higher the amount of data, higher is the deep learning architecture performance as compared to classical machine learning architectures.



Big data

Big data and artificial intelligence have a synergistic relationship. AI requires a massive scale of data to learn and improve decision-making processes and big data analytics leverages AI for better data analysis.

By bringing together big data and AI technology, companies can improve business performance and efficiency by:

- Anticipating and capitalizing on emerging industry and market trends.
- Analyzing consumer behavior and automating customer segmentation
- Personalizing and optimizing the performance of digital marketing campaigns
- Using intelligent decision support systems fueled by big data, AI, and predictive analytics

Python Packages for Machine Learning and Deep Learning

AI Big Data Analytics.

AI can assist users in all phases of the big data cycle, or the processes involved in the aggregation, storage, and retrieval of diverse types of data from various sources. These include data management, pattern management, context management, decision management, action management, goal management, and risk management.

AI can identify data types, find possible connections among datasets, and recognize knowledge using natural language processing. It can be used to automate and accelerate data preparation tasks, including the generation of data models, and assist in data exploration. It can learn common human error patterns, detecting and resolving potential flaws in information. And it can learn by watching how the user interacts with an analytics program, surfacing unexpected insights from massive datasets fast. AI can also learn subtle differences in meaning, or context-specific nuances, in order to help users better understand numeric data sources. And it can alert users to anomalies or unexpected patterns in data, actively monitoring events and identifying potential threats from system logs or social networking data, for example.

Machine Learning, as the name suggests, is the science of programming a computer by which they are able to learn from different kinds of data. A more general definition given by Arthur Samuel is – "Machine Learning is the field of study that gives computers the ability to learn without being explicitly programmed." They are typically used to solve various types of life problems.

In the older days, people used to perform Machine Learning tasks by manually coding all the algorithms and mathematical and statistical formulas. This made the processing time-consuming, tedious, and inefficient. But in the modern days, it is become very much easy and more efficient compared to the olden days with various python libraries, frameworks, and modules. Today, Python is one of the most popular programming languages for this task and it has replaced many languages in the industry, one of the reasons is its vast collection of libraries. Python libraries that are used in Machine Learning are:

- Numpy
- Scipy
- Scikit-learn
- Theano
- TensorFlow
- Keras
- PyTorch
- Pandas
- Matplotlib

Numpy

NumPy is a very popular python library for large multi-dimensional array and matrix processing, with the help of a large collection of high-level mathematical functions. It is very useful for fundamental scientific computations in Machine Learning. It is particularly useful for linear algebra, Fourier transform, and random number capabilities. High-end libraries like TensorFlow uses NumPy internally for manipulation of Tensors.

```
# for some basic mathematical
# operations

import numpy as np

# Creating two arrays of rank 2
x = np.array([[1, 2], [3, 4]])
y = np.array([[5, 6], [7, 8]])

# Creating two arrays of rank 1
v = np.array([9, 10])
w = np.array([11, 12])

# Inner product of vectors
print(np.dot(v, w), "\n")

# Matrix and Vector product
print(np.dot(x, v), "\n")

# Matrix and matrix product
print(np.dot(x, y))
```

utput:

```
219

[29 67]

[[19 22]
```

SciPy

SciPy is a very popular library among Machine Learning enthusiasts as it contains different modules for optimization, linear algebra, integration and statistics. There is a difference between the SciPy library and the SciPy stack. The SciPy is one of the core packages that make up the SciPy stack. SciPy is also very useful for image manipulation.

Scikit-learn

Scikit-learn is one of the most popular ML libraries for classical ML algorithms. It is built on top of two basic Python libraries, viz., NumPy and SciPy. Scikit-learn supports most of the supervised and unsupervised learning algorithms. Scikit-learn can also be used for data-mining and data-analysis, which makes it a great tool who is starting out with ML.

TensorFlow

It is a very popular open-source library for high performance numerical computation developed by the Google Brain team in Google. As the name suggests, Tensorflow is a framework that involves defining and running computations involving tensors. It can train and run deep neural networks that can be used to develop several AI applications. TensorFlow is widely used in the field of deep learning research and application.

```
#  Python program using TensorFlow
#  for multiplying two arrays

# import `tensorflow`
import tensorflow as tf

# Initialize two constants
x1 = tf.constant([1, 2, 3, 4])
x2 = tf.constant([5, 6, 7, 8])

# Multiply
result = tf.multiply(x1, x2)

# Initialize the Session
sess = tf.Session()

# Print the result
print(sess.run(result))

# Close the session
sess.close()
```

utput:

```
[ 5 12 21 32]
```

Keras

It provides many inbuilt methods for groping, combining and filtering data.

Keras is a very popular Machine Learning library for Python. It is a high-level neural networks API capable of running on top of TensorFlow, CNTK, or Theano. It can run seamlessly on both CPU and GPU. Keras makes it really for ML beginners to build and design a Neural Network. One of the best thing about Keras is that it allows for easy and fast prototyping.

PyTorch

It is a popular open-source Machine Learning library for Python based on Torch, which is an open-source Machine Learning library that is implemented in C with a wrapper in Lua. It has an extensive choice of tools and libraries that support Computer Vision, Natural Language Processing(NLP), and many more ML programs. It allows developers to perform computations on Tensors with GPU acceleration and also helps in creating computational graphs.

```python
import torch


dtype = torch.float
device = torch.device("cpu")
# device = torch.device("cuda:0") Uncomment this to run on GPU

# N is batch size; D_in is input dimension;
# H is hidden dimension; D_out is output dimension.
N, D_in, H, D_out = 64, 1000, 100, 10

# Create random input and output data
x = torch.random(N, D_in, device=device, dtype=dtype)
y = torch.random(N, D_out, device=device, dtype=dtype)

# Randomly initialize weights
w1 = torch.random(D_in, H, device=device, dtype=dtype)
w2 = torch.random(H, D_out, device=device, dtype=dtype)

learning_rate = 1e-6
for t in range(500):
    # Forward pass: compute predicted y
    h = x.mm(w1)
    h_relu = h.clamp(min=0)
    y_pred = h_relu.mm(w2)

    # Compute and print loss
    loss = (y_pred - y).pow(2).sum().item()
    print(t, loss)

    # Backprop to compute gradients of w1 and w2 with respect to loss
    grad_y_pred = 2.0 * (y_pred - y)
    grad_w2 = h_relu.t().mm(grad_y_pred)
    grad_h_relu = grad_y_pred.mm(w2.t())
    grad_h = grad_h_relu.clone()
    grad_h[h < 0] = 0
    grad_w1 = x.t().mm(grad_h)

    # Update weights using gradient descent
    w1 -= learning_rate * grad_w1
    w2 -= learning_rate * grad_w2
```

Output:

```
0 47168344.0
1 46385584.0
2 43153576.0
...
...
...
497 3.987660602433607e-05
498 3.945609932998195e-05
499 3.897604619851336e-05
```

Pandas

Pandas is a popular Python library for data analysis. It is not directly related to Machine Learning. As we know that the dataset must be prepared before training. In this case, Pandas comes handy as it was developed specifically for data extraction and preparation. It provides high-level data structures and wide variety tools for data analysis. It provides many inbuilt methods for grouping, combining and filtering data.

```
# Python program using Pandas for
# arranging a given set of data
# into a  table

# importing pandas as pd
import pandas as pd

data = {"country": ["Brazil", "Russia", "India", "China", "South Africa"],
        "capital": ["Brasilia", "Moscow", "New Delhi", "Beijing", "Pretoria"],
        "area": [8.516, 17.10, 3.286, 9.597, 1.221],
        "population": [200.4, 143.5, 1252, 1357, 52.98] }

data_table = pd.DataFrame(data)
print(data_table)
```

Output:

	country	capital	area	population
0	Brazil	Brasilia	8.516	200.40
1	Russia	Moscow	17.100	143.50
2	India	New Dehli	3.286	1252.00
3	China	Beijing	9.597	1357.00
4	South Africa	Pretoria	1.221	52.98

Matplotlib

It is a very popular Python library for data visualization. Like Pandas, it is not directly related to Machine Learning. It particularly comes in handy when a programmer wants to visualize the patterns in the data. It is a 2D plotting library used for creating 2D graphs and plots.

A module named pyplot makes it easy for programmers for plotting as it provides features to control line styles, font properties, formatting axes, etc. It provides various kinds of graphs and plots for data visualization, viz., histogram, error charts, bar chats, etc,

Python3

```
#  Python program using Matplotlib
# for forming a linear plot

# importing the necessary packages and modules
import matplotlib.pyplot as plt
import numpy as np

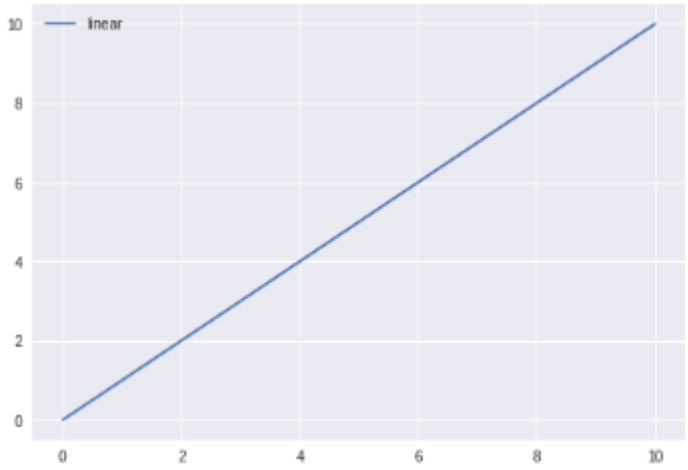
# Prepare the data
x = np.linspace(0, 10, 100)

# Plot the data
plt.plot(x, x, label ='linear')

# Add a legend
plt.legend()

# Show the plot
plt.show()
```

Output:



Visualization Library (**VL**) is an open source C++ middleware for 2D/3D graphics applications based on OpenGL 4, designed to develop portable applications for the Microsoft Windows, Linux and Mac OS X operating systems.